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Abstract

insert abstract

Acknowledgements

Generative AI Statement

In accordance with the IAS generative AI policy, this statement documents the use of generative AI tools in the preparation of this thesis.

A distinction must be drawn between two categories of AI use. First, large language models are employed as a research instrument within the methodology itself, as described in the methods chapter. This methodological use forms part of the empirical contribution of the thesis and is documented and validated there in detail.

Second, generative AI was used in a limited capacity as an aid for complementary, non-core tasks, in line with what the policy explicitly permits. Such tools were occasionally consulted to support coding work, debugging scripts and resolving syntax or tooling issues. All code was reviewed, tested, and adapted by the author, and the analytical decisions remain the author's own.

Generative AI was not used to produce the substantive text of this thesis. The framing, argumentation, theoretical engagement, methodological reasoning, and writing are entirely the author's own work.

Ethics Statement

The data underlying this thesis was collected in 2022 through the official Twitter API for academic researchers, such that no scraping or circumvention of platform restrictions was necessary and the platform itself was not adversely affected by the collection. All raw and processed data is stored locally and, where backups are required, on TU Dresden's secure Datashare; no commercial cloud service was used at any stage, and the LLM-based text analysis was conducted on TU Dresden's Capella HPC cluster using a locally hosted model, ensuring that the data never left the controlled research environments of MPI Leipzig, TU Dresden, and the author's local machine.

Ordinary users appear in the analysis only as constituent elements of aggregate engagement structures; no individual usernames or user-level identifiers from the general public are reported or otherwise made traceable.

Politicians, by contrast, are referred to by name, since their posting behaviour forms part of a public political discourse whose mechanics are of legitimate interest to the public they represent. This interest is not only morally defensible but also legally anchored in Article 40 of the Digital Services Act, which establishes platform-mediated political communication as a legitimate object of academic research on systemic risks (European Parliament and Council 2022).

1. Introduction

Social media threatens democracy through several channels, one of them being populism (Lorenz-Spreen et al. 2022). While scholars debate definitions, populist movements are democratizing authoritarian regimes and destabilizing Western democracies (Lorenz-Spreen et al. 2022). This is hardly surprising considering the essence of populism.

Populism research has historically been divided by geographical focus, methods, and host ideologies (Hunger and Paxton 2022). Since 2016 the research landscape has been dominated by political science perspectives converging on populism as a discursive practice (Tuğal 2021). They almost uniformly adopt Mudde (2004)‘s definition of populism as “an ideology that considers society to be ultimately separated into two homogeneous and antagonistic groups, ‘the pure people’ versus ‘the corrupt elite’, and which argues that politics should be an expression of the *volonté générale* (general will) of the people” (Mudde 2004:542).

While theoretical work refines the concept largely within Marxist traditions, relational and systems-sociological perspectives have remained peripheral (Hunger and Paxton 2022; Tuğal 2021). This is puzzling, because Mudde’s definition is itself relational: populism is constituted through the antagonism between two groups. Simmel already connected antagonism to group cohesion, noting that groups are “...held together by a shared aversion [...] to which entirely foreign elements are drawn by the commonality of an antagonism.” (Simmel 1908:Ch.4). A relational reading of Mudde is therefore not a far fetched but literally a theoretical operationalization of it.

One of computational social science’s original aims is to make use of digital data for social science questions (Lazer et al. 2009). Since the dynamics of complex systems are baked into the design of social media, researchers should take a complex-systems perspective when studying them (Bak-Coleman et al. 2025). Yet existing large-scale observational social media studies of populism analyze only textual content (Erhard et al. 2025; Serrano, Papakyriakopoulos, and Hegelich 2020; Yarchi, Baden, and Kligler-Vilenchik 2021). Populism has rarely been connected to networks.

This study fills that gap by asking:

“How does populist rhetoric by German parliamentary members shape the structure of their reply communities on Twitter?”

Populism research is divided over whether populism is a thin or a thick ideology. The minimal definition reads it as a vertical antagonism between a homogeneous people and a homogeneous elite (Mudde 2004), often enriched with a horizontal dimension constructing the people additionally against outside groups (Brubaker 2017). What unifies all perspectives is the minimal definition as a common denominator and the intuition that populism is system-destabilizing (Lorenz-Spreen et al. 2022). The definitions of populism implicitly are connected to representative democracy. Democracy selects its communications through the Government/Opposition distinction (Luhmann

1987), and populism is then the reproduction of this distinction at the level of a social movement (Elite/People) attempting to place itself as a valid democratic group. Social media amplifies this code (Lorenz-Spreen et al. 2022) and Twitter in particular is not a neutral arena in which rhetoric is received but a medium whose recommender architecture actively structures who encounters whom (X 2026). Parties are then movements competing for political power in this attention environment (Simon 1971; Tilly 1978:117), and populist rhetoric constructs the distinction of the pure people against a corrupt elite. This gives encounters a distinction around which they can stabilize. Engagement in this medium is not agreement (Morselli et al. 2026) but a structural fact, users repeatedly showing up in the same place around the same politician whatever their reason for doing so, and whether such stabilization actually occurs is then a question about communication and not about speakers.

The methodological approach translates the outlined theoretical ideas by using a twitter reply network to operationalize social interactions on social media. Compared to retweets and mentions replies are a way of directly answering to a message another person published. The dataset contains politicians tweets and their replies to construct such a reply network. To explore the effect of a politician using populist rhetoric on their community all tweets are labeled by the amount of populist attitude they communicate. This textanalysis is achieved through a fewshot LLM Qwen3-235B annotation using an expert validated systemprompt. The egonetworks of politicians are extracted and their structure compared by their usage of populist rhetoric to explore if the expected relationship can be observed.

- Results & Discussion

2. Literature Review

Populism research is divided over one main question: Is populism a thin or thick ideology? And if it is thick, how so? The following section provides an overview of the conflict and explains the implicit theoretical source of the conflict. The second half proposes a new generalizing perspective on the topic motivated through already observed micro mechanisms.

2.1. Populist Theories of Populism

2004 Cas Mudde proposed a minimal definition of populism as an ideology where a message is framed in a way that the powerless homogenous people stand against a ruling homogenous elite as antagonistic groups (Mudde 2004). This definition has been considered “thin”, because it only considers a vertical divide of society (Brubaker 2017). Vertical in the sense that one group is ruling the other. When read as an ideology Mudde (2004) s mininmal populism implicitly includes a class component as the core of its definition (Yates and Mondon 2026). The definition is currently the most cited work on populism. This can be explained with the minimalism increasing the potential of connected communication for theorists discussing the work (Brubaker 2017; Luhmann 1987; Tuğal 2021; Urbinati 2019). Empirical studies operationalized the main dimensions of people-centrism and anti-elitism in surveys and finetuned bert classifier PopBert on german Bundestagspeeches (Akkerman, Mudde, and Zaslove 2014; Castanho Silva et al. 2020; Erhard et al. 2025). Often this minimal definition is combined with some other dimension like antipluralism or the distinction between left and rightwing populism (Castanho Silva et al. 2020; Erhard et al. 2025). Theorists combine the thin definition with institutional and substantive analyses when discussing populism but those are chosen rather arbitrary (Tuğal 2021). The minimal definition is part of the same family of subjectivist definitions as discursive and performative marxist and postmarxist definitions built on Laclau (2005); Tuğal 2021.

Combining the minimal definition with more substantial categories of analysis not only thickens the analytic concept but also increases its theoretical richness. Brubaker (2017) criticised the minimal definition for only capturing his proposed vertical dimension without acknowledging the horizontal distinctions between “the people” and outside groups. For example one thickness-increasing addition would be that left-wing populism defines “the people” economically or politically against threats like globalization and imperialism, while right-wing populism defines them culturally or ethnically against outside groups and “internal outsiders” perceived as not belonging to the nation (Brubaker 2017). Engesser et al. (2017) use this thick definition to explore how they are used in a social media context and found that in Facebook and Twitter posts not one post contained all dimensions but that populist ideology was “fragmented” across posts. The more terms are added and connected the stronger and thicker the assumed ideology gets. Brubaker (2017) for example researches 5 different elites, more outsidegroups and

the heartland. This comes with the caveat that the more the concept of populism is enriched the more it “lumps together” disparate political projects with disparate social bases and modes of action (Brubaker 2017; Medzihorsky and Lindberg 2024).

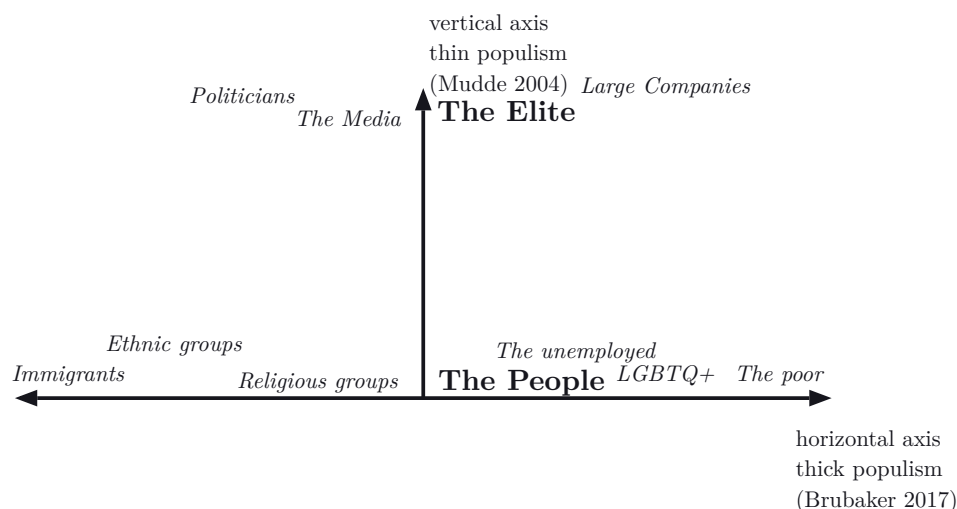


Figure 1: Vertical and horizontal dimensions of populism

After Mudde (2004) and Brubaker (2017). The vertical axis captures the thin, minimal definition: “the people” constructed against “the elite”. The horizontal axis captures Brubaker’s thick extension, in which “the people” is additionally constructed against horizontal others. The labels along the axis are illustrative examples of such horizontal others; their position on the axis is not meaningful and the horizontal dimension is not equivalent to a left–right axis.

But is populism really such a densely connected set of concepts, aka an ideology? And is it democratic or antidemocratic? While those putting left and rightwing populism on one axis do not necessarily assume that they are part of the same dimension they use it to enrich populism in order to bring it closer to an ideology. Yates and Mondon (2026) argue that on an ideological level there is no right-wing populism. Leftwing and rightwing populists disagree on the meaning of the “people” they invoke. The leftists say “plebs”, the rightwing say “ethos” when saying “the people”. While left wing populism aims at increasing marginalized groups visibility, right wing populism extends the privilege of already visible groups and leaves the oppressed and exploited oppressed and exploited (Yates and Mondon 2026). For rightwingers populism is then not part of an ideological core but merely a strategy for gaining votes. They abuse populisms ability to unify different cleavages splitting the critique of elites for different reasons (Urbinati 2019). In short, even though thin vertical populism is often enriched with the left and right dimension only leftwing populism can be ideological. The distinction becomes clearer when populists gain power. Populist leaders in power either (1) reaffirm their pro-people identity, remaining in a permanent electoral campaign or (2) change rules to strengthen their decisionmaking power (Urbinati 2019). While the populists continue to use propaganda facist populists revoke checks and balances (Urbinati 2019). Populist ideology arrives at a so called “partyless democracy” where the people represent themselves through populist protests and parties openly rule only for their own good (Brubaker 2017; Kriesi 2014; Urbinati 2019).

Luhmanns Systemtheory provides a contrasting image to the relation between populism and a democratic state of the different scholars. For Luhmann democracy is not making all decisions participatory because then one would reduce all decisions to decisions about decisions which Luhmann calls “Teledemobureaucratization” (Luhmann 1987). This favours opaque powerstructures and insiders, just like the partyless democracy claims. In its essence democracy for Luhmann is the division of the elite into government vs. opposition which becomes the systems binary code to select which communications are considered relevant or not. What is so special about this binary code in particular is that it dissolves fundamental paradoxes that all systems with organized powerdifferences inherit (Luhmann 1987). This split only works when society is already differentiated in enough horizontal functional systems that it does not need a head of state anymore (Luhmann 1987). Society then is too big and previously relevant systems have emancipated themselves from their role, sustaining themselves through autopoiesis. Descriptively, this overlaps with the diagnosis of the powerless democracy, even if Luhmann’s framework remains value-neutral where Urbinati’s is normative. Crucially though, “elite” here is a functional designation, those holding or contesting office, not a moral indictment. Populism’s elite/people distinction is something else: it reintroduces a moral coding, pure vs. corrupt, into a political system whose code is government vs. opposition. From a Luhmannian perspective the thin populist ideology therefore does not naturally arise from the code but presses against it, as an attempt to override the functional split with a moral hierarchy.

At this point a distinction between two cases can be made, although both already presuppose a representational vocabulary that Luhmann himself tries to dissolve. Either the representative democracy is seen as a genuine and working representation of the people or it is seen as a functionally differentiated system that emancipated itself from society where parties already only rule in their own favor. In the first case representative democracy is the rule of the people and in the second case representative democracy is the rule of the elite. Read through Luhmann the second framing is closer to the operational reality, and the thin definition is then an accurate representation of the actual structural condition where thickening would be optional.

When scholars choose between a thin or thick definition of populism, they often commit implicitly to a perspective on democracy, even where the choice is presented as merely methodological. A thick definition adds content like ethnic identity, antipluralism, or outside groups, and it treats populism as something that corrupts a working democracy. This only makes full sense if you already believe democracy genuinely represents the people, otherwise there would be nothing for populism to corrupt. A thin definition, with just the elite-people divide, fits much better with the idea that democracy has closed in on itself and mostly serves its own elite. The thin divide then describes a real structural split rather than a distortion. Most researchers overlook this tension, and some hold both sides at once, they define populism thinly but still frame it as a danger to a democracy that genuinely works. Read through Luhmann, populism is at once an accurate diagnosis of a political system that emancipates itself from the people it wants to represent and a reactive moralization at the same time that would collapse

the government/opposition code by declaring one pole legitimate and the other not. The thin definition is therefore revealing in both directions. It captures the structural condition the powerless-democracy describes and it carries within itself the moral re-entry that makes populism a de-differentiation threat rather than a corrective. In other words, populism describes a real problem in the political system as a differentiated functional system, but its moralizing it at the same time trying to replace the structural code with a moral one.

What in the end unifies all perspectives is the minimal definition as a common denominator and the general intuition that populism is political-system-destabilizing (Lorenz-Spreen et al. 2022).

2.2. Populism and Social Media

Social media populism studies all implicitly understand populist as a form of communication since they analyse it on a social media platform connecting all of its users.

If populism arises from the elite-people code of representative democracy, social media amplifies this code and decouples it from traditional institutions. Digital media use is consistently associated with higher populism, with causal evidence for far-right support and even ethnic hate crimes in democratic and authoritarian regimes (Lorenz-Spreen et al. 2022). The question is not whether social media matters but what it does structurally. The common explanation is affordance. Social media is cheap, viral and engagement-driven, exploited by populists for demagogic, anti-establishment and people-praising messaging including name-and-shame strategies (Gil De Zúñiga, Koc Michalska, and Römmele 2020). Cassell (2021) confirmed this through qualitative coding of populist leaders tweets across Latin America and Europe, where populist frames outperformed pluralist, technocratic and neutral tweets in likes and retweets. Yet this engagement advantage is asymmetric. Dutch populists reciprocate interaction on Twitter less than non-populists (Jacobs and Spierings 2019). Populist communication mobilizes attention upward without distributing it downward. Politicians also adapt to platforms rather than to policy. Stier et al. (2018) used a Bayesian semi-supervised single-membership language model to show that candidates use Facebook and Twitter for distinct purposes, discussing campaign events and platform-specific topics rather than policy. The audience is no longer a mass but chooses its broadcaster, which makes strategic tailoring rational. Hu (2024) explores the engagement of political tweets labeled by topics and finds few follower of politicians using selfexpressive, argumentative as well as mobilizing language. These patterns aggregate at the group level. Stier et al. (2017) showed AfD and Pegida share Facebook userbases, mutually like each other and converge on topics like crime, sexual assaults, EU referenda and “state and the people”. Stier, Siegers, and Breuer (2025) found through linked survey and webtracking data that radical right populists and their supporters avoid public broadcasters and expose themselves to alternative channels. “How to talk about and select media sources seems to have developed into a core component in the construction of a radical right group identity and a shared information ‘safe space’ for political information” (Stier et al. 2025).

Group identity is built around the choice of information infrastructure itself. This ties populist communication to a legitimacy crisis rather than to a left-right axis. Analyzing 32 million tweets from parliamentary accounts in 26 countries, Törnberg and Chueri (2026) found that neither left-right nor populism alone explains misinformation spread. Both left and right-wing populists consume more misinformation, but only the rightwing believe themselves better informed. The radical right has built an alternative media ecosystem in symbiotic relationship with attention-economic platforms (Törnberg and Chueri 2026). Consistent with the Luhmannian reading above, once a functional system emancipates itself from its societal task it begins to produce its own environment to keep itself alive.

To summarize, populism is a style of communication that constructs the divide between the people and the elite with usually rule-destabilizing consequences. Even though we can not know the individuals motivations or reasons behind usage of populist rhetoric, what is sure is the role politicians assign themselves to and what being a politician implicates. From a point of political communication politicians use this language in order to put themselves into a position to mobilize and govern.

2.3. Language and the Meso Level

Up until now it was established that populist language is used by politicians on social media platforms and that it has been connected to certain language aswell as engagementmetrics.

But engagementmetrics only report on the reactions of single users, but users interact, especially on social media platforms. Those interactions of single users are conceptually not independent since social media recommender algorithms use collaborative filtering. This is especially well known for Twitter since X open sourced the architecture of the platforms recommender algorithm (X 2026). Multiple steps in the recommender algorithm are enriched with collaborative filtering techniques like SimClusters, the UTEG (User-Tweet-Entity-Graph) and the knowledgegraph (X 2026). Through this algorithm Twitter shapes the interactions on its platform. But not only from a technical perspective groupdynamics between users should be considered. One of the oldest sociological idea is that antagonistic groups shape each other. “... that through [the dispute] not only does an existing unit concentrate itself into a more energetic unit, and radically eliminate all elements that could blur the sharpness of its boundaries against the enemy – but that [the dispute] brings together persons and groups who otherwise had nothing to do with each other.” (Simmel 1908:251).

For Simmel a group needs at least three persons (Simmel 1908:Ch2). This is mainly because when one person leaves the group it doesnt automatically dissolve like if in the case of only two persons. For three persons the group can sustain itself if a person leaves the group. That a group emerges in the real world two processes are needed, homophily for the group cohesion and repulsion of others for defined group boundaries (Stadtfeld, Takács, and Vörös 2020). Pure attraction only explains a groups

expansion while a heterophob repulsion mechanism creates stable groups by defining its boundaries. Stadtfeld et al. (2020) show this by calibrating a stochastic actor oriented model to 479 students from 13 schoolclasses in order to simulate how friendship and dislike networks emerge. In Twitter retweet networks language is used as a marker for opinion-based group formation (Morselli et al. 2026). Opinion based group formation requires opinions as identity markers that individuals use to transition from holding an opinion to selfcategorizing themselves through the opinion (Morselli et al. 2026). In linguistic theory the identity markers changing meaning for the ingroup is described through the distinction between esoteric and exoteric language (Wray and Grace 2007). Esoteric language is specialized in the sense that it is used for ingroup communication and outsiders can not understand it. Exoteric language is more selfexplanatory and is used for communication with outsiders (Wray and Grace 2007). The coevolution of language and social groups is already actively researched in the science of science. For example Schmitz, Schmidt-Wellenburg, and Volle (2025) trace the evolution of scientific groups in the US and German sociology through a stochastic blockmodel of a multilayer network operationalized through co-word usage, shared citations and co-authoring.

The social media studies so far were mostly largescale observational studies. Experimental studies in controlled contexts can give closer insight into the social mechanics of online group/discussion formation. Oswald, Schulz, and Lorenz-Spreen (2025) payed participants to engage with each other about political topics in Reddit forums they moderated. Participants additionally filled out surveys regularly over a period of four weeks. When users percieved a discussion to be toxic they did not engage in the discussion. The users who engaged, engaged more when the discussion was polarized and toxic (Oswald et al. 2025). With a Luhmannian read the results can be generalized to the formation of a code in the discussion as a social system. Users percieve the community as positive/negative according to their own standards and start with the second stage of communication: “Mitteilung” (Luhmann 1984:Ch.4). While different persons engage according to their own evaluative codes their communication creates thematic and meaning structures (Luhmann 1984:Ch.4). The social-discussion-system starts to reproduce itself. If now the discussions would start to be selfdescriptive the system starts making a distinction between itself and its environment and emerge as a differentiated unit itself (Luhmann 1984:Ch.4). Oswald et al. (2025)s experiment also include the persons being driven away from the discussion through e.g. the evaluation of a toxic environment. With the Luhmannian read, outside of the experimental context they would put their attention somewhere else, creating new opportunities for social systems to emerge, driving again others away. On one hand the discussion becoming more homogenous reduces complexity, but it also increases comprehensible complexity Luhmann (1984). Higher connectivity (Netness) around a certain category (Catness) is equivalent to a higher amount of organization of a group (Tilly 1978:63). Parties are aggregating members loyal to the category, aka political interest and communicative practice, aka code, making it distinct from its environment (Tilly 1978:76). Parties are then social movement because they are mobilizing others (Kusche 2016; Tilly 1978).

Intergroup violence is not motivated by the out-group hate but through the spiraling “cheap talk” happening inside the in-group (Mäs and Dijkstra 2014).

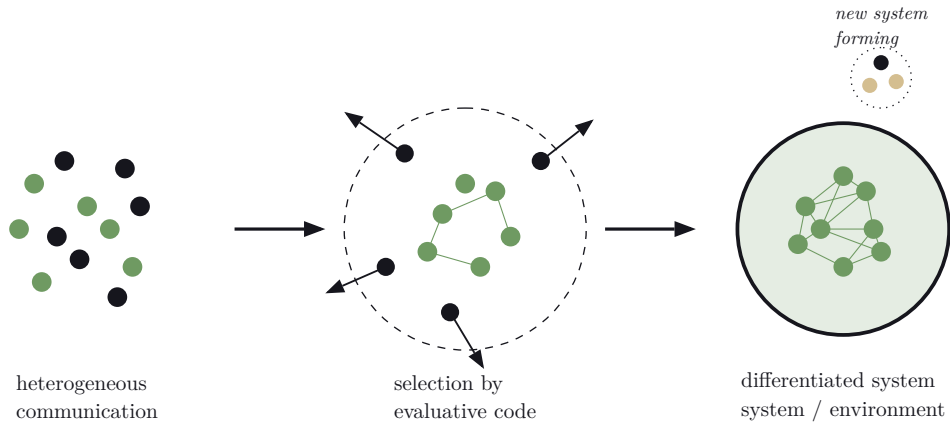


Figure 2: Three-stage account of group emergence in online discussions

After Oswald et al. (2025) read through Luhmann (1984) and Tilly (1978). At the first step communication is heterogeneous: users with divergent evaluative codes (positive in green, negative in pink) all participate in the same discussion. At the second step the system selects: users who evaluate the discussion negatively disengage, while those who evaluate it positively intensify and form initial ties. At the last step a homogenized code reproduces itself, ties densify (Tilly’s netness around a shared catness), and the discussion makes a distinction between itself and its environment, emerging as a differentiated social system. The displaced users seed new systems elsewhere, restarting the cycle.

Populism (Elite/People) is the reproduction of the democratic system (Government/Opposition) through the social movement as a subsystem of the political system. Democratic system select communication through this code (Luhmann 1987) and by reproducing it the social movement attempts to place itself in a position of being a valid democratic group. This explains why right-wing populists use this distinction as a mobilization strategy even though it contradicts their elitist ideology (Yates and Mondon 2026). Being for the people as an ingroup is very universal since every voter can be part of the constructed ingroup. Communication with this distinction has great potential for “Anschlusskommunikation”. It drags everyone in the political arena, puts citizens at the negative-opposition side of the Government/Opposition distinction inviting them to participate in the movement empowering them as political citizens.

A more active engagement community should therefore manifest itself through more observed ties in the area around the populist politician.

Or more precisely for the given Dataset:

German MPs who use populist rhetoric have higher interconnected alters in their reply ego networks compared to MPs using less populist rhetoric.

Engagement networks are special in the sense that they do not reflect who agrees with whom but who interacts with whom. Morselli et al. (2026) found that close individuals in the twitter retweet network can have different opinions. Even though on a linguistic basis at the first glance it seems unreasonable, but we can imagine the extreme case. A user is always coming back to a politicians tweets and its fanbase to share their

disagreement. Are they part of the group or not? The person is not part of the fanbase but definitely part of an engagement community where individuals continuously show up in the same place and position, no matter why.

3. Data and Methods

This section describes the data, measures, and analytical strategy used to examine how populist rhetoric by German MPs shapes their engagement communities on Twitter. After introducing the dataset and collection procedure the 3 step analysis is described. First, as the main part of a text analysis populism is operationalized as a three-dimensional construct of People Attitude, Elitist Attitude, and Antagonism, scored by a large language model through a structured annotation prompt. Second, the construction of reply threads, the user-level reply network and the exploration of their structure are detailed aswell. Finally in a synthesis, politician ego networks are combined with the populism scores produced by the text analysis. Linear Regressions with their controls, aswell as a robustness check and further exploration of the ego networks is described.

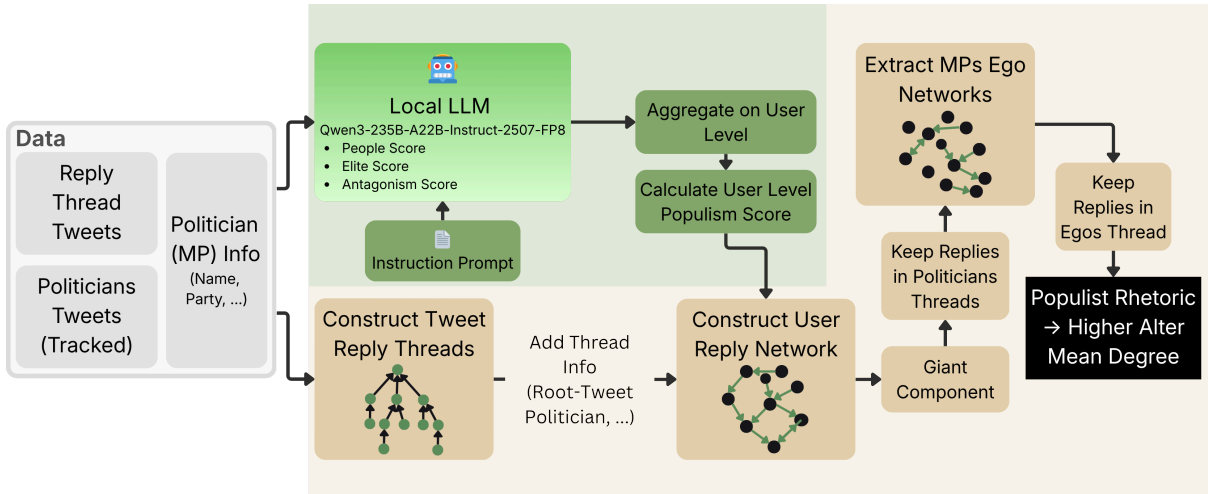


Figure 3: Diagram describing conducted Analysis

3.1. Data and Operationalization

The dataset used in this study was kindly provided by Armin Pournaki from the Max Planck Institute for Mathematics in the Sciences, Leipzig. The dataset holds tweets and replies directed at German members of parliament aswell as their retweets and referenced tweets, collected over a one wekk from February 7 to February 14, 2022. Only the MPs tweets and their replies are analysed since the reply network best captures social interactions of engagementcommunities. Data were obtained via Twitter's streaming API using two parallel strategies. The follow stream tracked replies to MP-generated tweets, while the track stream captured tweets mentioning MP handles (e.g., "@username"), including direct and nested replies. Protected accounts were excluded. Both datasets were combined and deduplicated. From the raw data, reply-threads were constructed by linking tweets through directed reply chains. These interactions were then aggregated to a user-level reply network, weighted by the number of exchanges. Thus, the dataset enables an analysis of interaction patterns between politicians and

their engagement communities on social media. To delineate the set of politicians and add their name and party information, a list of parliamentary members of the 19th, 20th, and 21st German Bundestag is obtained from the Bundestags Webarchiv and added to the original dataset through their twitter-account links (Bundestag 2026). The complete dataset consists of 693 015 Tweets in the twitwi format (medialab 2026).

German political news from 7–14 February 2022 were dominated by the Omikron wave’s peak and the escalating Russia-Ukraine crisis. As Russia massed troops for what would become Europe’s largest military offensive since WWII, Chancellor Scholz met President Biden to discuss Germany’s Nord Stream 2 dependency. Domestically, Scholz’s coalition split over a general vaccine mandate, with FDP Justice Minister Buschmann deeming it constitutionally dubious and proposing mandatory physician consultations for unvaccinated adults as a softer alternative.

Current studies almost uniformly base their understanding of populism on Mudde (2004)’s definition of populism as “two homogeneous and antagonistic groups, ‘the pure people’ versus ‘the corrupt elite’, and which argues that politics should be an expression of the *volonté générale* (general will) of the people.” (Mudde 2004:p.543). While all operationalizations include a “pro-people” and “anti-elite” one of multiple third dimensions is often implemented as well like the inclusion of anti-pluralist attitudes, the distinction between leftwing and rightwing populism or agitating against horizontal outgroups like minorities (Aalberg et al. 2017; Castanho Silva et al. 2020; Meyer et al. 2025). Populist Attitudes are not only measured through surveys (Castanho Silva et al. 2020), but also through observational studies of political discourse on social media (Meyer et al. 2025) leveraging LLMs.

This study operationalizes the core dimensions of Mudde (2004)’s definition, People Attitude, Elitist Attitude, and Antagonism, by instructing a large language model through an annotation prompt (see [Appendix](#)). Each dimension is defined with explicit scoring anchors: People Attitude and Elitist Attitude are measured on bidirectional scales from -3 to $+3$, where positive values indicate support for and negative values indicate opposition to the respective group, while Antagonism is measured on a unidirectional scale from 0 (no divide) to 6 (existential threat), with labeled thresholds distinguishing dissatisfaction (1–2), active blame (3–4), and existential threat framing (5–6). The prompt leverages chain-of-thought style reasoning and few-shot examples to guide the models annotation behaviour. To guide the models reasoning letting it question itself throughout the process lead improved the results immensely compared to hard rule-based checks.

Calculating populism as a composite is necessary ...

- clarify combining dimensions
 - just looking for anti elite is anti elitism
 - just looking for pro people does not account for the anti elite dimension part of the literature, additionally 2 more arguments. Why pro people is not enough. maybe the group delineation is okay to include

here?? Being for the ingroup might radicalize, but is not enough for clear delineation

- ▶ Refer to figure 1
- and why summing/multiplying up is done
 - ▶ populism is fragmented (Engesser et al. 2017)
- why not added ==> because it doesn't make sense for antagonism
 - ▶ why do I need antagonism? because the xyz is stronger
 - ▶ internal validity is unimportant, it comes from psychometrics and assumes all elements of the scale measure the same thing. here this is not the case

The prompt follows the structure described by Liu et al. (2026). It begins with a role definition and a pre-analysis check if the text carries any content other than just a link or a user mention. It proceeds by defining “People Attitude”, “Elite Attitude” and “Antagonism”. The definition of “the people” is restricted to a broad ordinary majority and explicitly excludes named individuals, lists of specific persons, and narrow subgroups unless the text frames them as standing in for the general public. Similarly, elite criticism is only scored when the target is a generalized powerful class rather than a single individual or a specific policy disagreement. In the last major section the prompt invokes the chain-of-thought before assigning scores, the model must produce a holistic redescription of the post’s rhetorical strategy, an actor-by-actor analysis that classifies each referenced person or group by scale (individual, institution, or generalized class) and dimension-specific explanations that articulate the reasoning behind each score. At each step, the model is asked to consider alternative readings and flag its confidence as LOW when a reasonable coder could disagree. Three few-shot examples are included to calibrate the model’s decision boundaries: a strongly populist post with high people, elite, and antagonism scores, a non-political post that should receive all zeros, and an ambiguous post where institutional criticism could plausibly be read as either targeted policy dissatisfaction or broader anti-elite attitude. This example structure is designed to discourage binary classification tendencies and encourage the model to use the full range of each scale.

The prompt is included as a systemprompt and appended with the to be annotated tweet. The model outputs its reasoning and scores in a json format.

After coding the scores are combined. The three dimension scores are denoted $P \in [-3, +3]$ (People Attitude), $E \in [-3, +3]$ (Elitist Attitude), and $A \in [0, 6]$ (Antagonism). The tweet-level populism score is

$$\text{Populism} = \begin{cases} (P - E) \times A & \text{if } A \geq 1 \\ P - E & \text{if } A = 0 \end{cases}$$

The subtraction captures the joint rhetorical direction that is positive when the people are elevated and elites denigrated yielding a theoretical range of $[-36, +36]$. Politicians using populist strategies on social media do not have to address all populism dimensions in one tweet but can be anti-elitist in one and pro-people in the next. Therefore the

combination on the user level is achieved by averaging each dimension separately before recombining. Let $\bar{p}_u$, $\bar{e}_u$, and $\bar{a}_u$ denote the per-user means. The user-level score is

$$\text{Populism}_u = \begin{cases} (\bar{p}_u - \bar{e}_u) \times \bar{a}_u & \text{if } \bar{p}_u > 0, \bar{e}_u < 0, \bar{a}_u > 0 \\ \bar{p}_u - \bar{e}_u & \text{if } \bar{p}_u > 0, \bar{e}_u < 0, \bar{a}_u = 0 \\ 0 & \text{otherwise} \end{cases}$$

The gate zeros out users whose average rhetoric is not simultaneously people-affirming and elite-critical, operationalizing Mudde (2004)‘s tripart definition at the actor level. Aggregating dimensions before recombining allows a user to distribute their anti-elite and pro-people attitudes across different tweets.

3.2. Textanalysis

Computationally methods driven text analysis is based on linguistic concepts. A corpus of documents is divided in tokens as semantically meaningful units of analysis which are often words. The simplest text classification techniques are usually wordfrequency based dictionary methods. Some of which already mechanistically take context of wordappearance into account (Ribeiro et al. 2016). While simple machine learning classifiers like naive bayes models are statistically more elaborate, deeplearning models are able to integrate interactions between tokens allowing for even better predictions. But deep learning architectures like convolutional neural networks trade variance for increased local bias through usage of filters (Sohil, Sohali, and Shabbir 2022). Transformer architectures implement modeling of long-range dependencies by implementing self attention on a micro level weighing the importance of every individual input token for other each individual input token (Cordonnier, Loukas, and Jaggi 2020). Qwen3-235B-A22B-Instruct-2507-FP8 is using more elaborate attention mechanisms like grouped query attention (Yang et al. 2025). The model also implements expert segmentation, is multilingual and has 8-bit floating weights (Yang et al. 2025). To run the model most efficiently on the HPC Capella Cluster the vLLM framework was used through python to make use of virtual memory PagedAttention techniques (Kwon et al. 2023).

Output quality of large language models is highly dependent on prompt design (Liu et al. 2026). Prompt engineering is conducted iteratively by drawing small random samples of tweets from the corpus and classifying it using an initial prompt. The resulting reasoning steps contained in the LLMs JSON output are then qualitatively assessed with regard to coherence and later label accuracy. Based on identified shortcomings, the prompt is systematically adjusted and the process is repeated. Early versions of the prompt contained were structured in the style described by Liu et al. (2026), later additional rule based checks were integrated, the coherence of the phrasing was adjusted and lastly the rulebased checks were changed to questions prompting the LLM to question its judgements and certainty. External validity is tested after improving the prompt qualitatively until reaching satisfying reasoning performance. The expert annotated training dataset of Bundestag for fine tuning the PopBERT populism classification transformer model is used (Erhard et al. 2025). The dataset contains 8795 sentences of

Bundestag speeches with binary annotations of five experts on the people-centrism and anti-elite dimension (Erhard et al. 2025). After annotating validation dataset with the developed systemprompt and Qwen3-235B-A22B-Instruct-2507-FP8 accuracy

$$\text{Accuracy} = \frac{1}{n} \sum_{i=1}^n \mathbb{1} \left[\text{rater}_A^{(i)} = \text{rater}_B^{(i)} \right],$$

aswell as F scores

$$\text{Precision} = \frac{\text{both rated positive}}{\text{both rated positive} + \text{only predicted positive}}$$

$$\text{Recall} = \frac{\text{both rated positive}}{\text{both rated positive} + \text{only reference positive}}$$

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

for pairs of expert raters with each other and the LLM annotations are calculated. For validation the bidirectional rating scale of the LLM is collapsed to binary labels matching the direction of pro-people and anti-elite. While the F1 score is not symetric for swapping labels it is symetric for swapping raters (see [Appendix](#)).

3.3. Network Analysis

The reply threads are constructed by treating tweets as nodes that are linked through directed replies. A component in the constructed thread network is then a replythread with one rootnode that is only recieving links marking the direction of a reply. Some reply chains trace back to MP tweets posted before the collection period, so the root node has no row in the dataset. Matching the target tweets user information against the politician table recovers their authorship, recovering 2,631 of 89,561 threads (2.9%). Threads root tweet information like politician with e.g. party and followerinformation, are then stored within the reply tweets of the respective thread and aswell added to each tweet in the reply network.

The main research question requires a network of interactions plausibly representing an engagement community. A weighted user centric reply network with users as nodes and replies as directed links is constructed. The original tweet-based dataset holding politician and threadinformation served as the basis for construction with its `user_id` and `to_userid` variables. The resulting network is directed with edgeweights based on replycount. It holds 81 295 user-nodes connected through 239 502 reply-links distributed among 1865 components. A giant component holds (77 194) 94% of all nodes. 1207 users are contained in components of size 2, one original user, one replying user, as the threadsize holding the second most nodes. The second largest component held 41 (0.05%) users (see the [table](#) in appendix). The network analysis is based on the giant component to make computation more feasable leading to a loss of the network periphery without any actual structurally relevant threads.

Tweet-level populism dimensions (people scores, elitism scores, antagonism scores) are aggregated per user by computing weighted means across each user’s tweets, then combined into a userlevel composite populism score. These user-level scores are added as vertex attributes to the reply network.

To definitely deliniate the dataset to replies from politicians tweets reply-edges not appearing underneath politicians threads are deleted and isolates are removed. Thereby removing rest-replythreads of e.g. referenced tweets.

3.4. Ego Network Analysis

A descriptive exploratory analysis describes first the results of the textanalysis and second the networkstructure. **its all graphbased, except the partyplot and the tweet-topics plot** The textanalysis is conducted with tweets contained in the largest component of the retweet network, that also serves as the basis for the network analysis. Exceptions are the comparison of populism by parties and a wordcorrelation network to infer the tweets content, which are based on the entire tweetdataset including retweets and referenced tweets in order to capture the broader political discourse during the week.

Following that the reply network structure is explored through a visualization using the Distributed Recursive Graph Layout and validated through closer inspection of degree and local clustering distributions (Martin, Brown, and Wylie 2007). The Distributed Recursive Graph Layout (DrL) lays out a graph by applying repulsion and attraction forces between nodes to prevent overlap and keeping connected nodes close. Beginning with randomness so nodes can move freely and avoid poor configurations, gradually reducing movement until positions stabilize. Once positioned, spatially close nodes are merged into representative summary nodes producing a coarser version of the graph. The cycle of coarsening and repositioning repeats until the graph is sufficiently small to lay out directly. Finally the process is reversed, expanding each simplified graph back one level at a time, using the prior layout as an initial arrangement.

The structure of politicians ego networks is compared to answer the main question about local engagement communities. Mean alter degree is used as the primary measure of alter interconnectedness, capturing the average number of connections each alter maintains to other alters within the ego network. While density would be a more established measure in usual cases it is inversely related to network size, making comparisons across ego networks of different scales unreliable. As a robustness check, the fragmentation ratio is additionally tested since this measure relates the number of connected components to the overall network, providing a complementary perspective on whether alters form a cohesive neighborhood or fragment into isolated clusters.

For controlvariables the politicians (ego) degree and followercount control for the popularity of a politician while mean thread size of the politicians seeded threads controls for the depth of conversational engagement elicited by that politician. Ego degree captures structural prominence in the reply network, followercount proxies platform-level visibility, and mean thread size accounts for the possibility that longer threads

mechanically increase the chance of alter-alter interaction simply by providing more opportunities for reply.

Two OLS regression models are estimated. The baseline model regresses mean alter degree on a binary populism indicator derived from the composite populism score. The full model adds ego degree, follower count, and mean thread size as controls. Comparing the populism coefficient across both models reveals how much of the observed relationship between populist rhetoric and alter interconnectedness is attributable to confounding structural and visibility differences between politicians.

As a robustness check, the fragmentation ratio

$$\text{Fragmentation} = \frac{\text{component count}}{n_{\text{alters}}}$$

is tested as an alternative dependent variable using the same specification as the full model. Where mean alter degree captures the intensity of alter interconnectedness, fragmentation captures its inverse: the degree to which the ego network consists of isolated clusters. A consistent result across both measures strengthens the claim that populist rhetoric is associated with more cohesive engagement communities instead of an artifact of the chosen operationalization.

To further shed light on which social mechanisms explain a difference in alter interconnectedness, observed edge statistics are compared with a random baseline within each politician’s ego network. For every ego network, three alter-level attributes are considered: follower count, tweet count, and the composite populism score. Follower count and tweet count proxy platform visibility and activity level, testing whether tie formation is driven by status homophily or preferential attachment among high-activity users. The composite populism score tests the substantively central mechanism: whether alters under populist politicians connect along ideological lines, forming ties with others who express similar levels of populist rhetoric.

Two ERGM-inspired statistics are computed for each attribute.

The absolute-difference statistic captures homophily. For every connected alter pair the mean absolute attribute difference $\overline{|x_i - x_j|}_{\text{obs}}$ and for every possible alter pair $\overline{|x_i - x_j|}_{\text{rand}}$ are calculated. A negative relative deviation signals that connected alters are more similar than chance would predict.

The sum statistic captures preferential activity. The mean attribute sum across connected pairs $\overline{x_i + x_j}_{\text{obs}}$ is compared to the mean across all possible pairs $\overline{x_i + x_j}_{\text{rand}}$, with positive deviation indicating that edges disproportionately link high-attribute alters. Both statistics are expressed as relative deviations from the random baseline $(o - r)/r$ and summarized by median and mean across populist and non-populist ego networks to identify possible systematic group-level patterns in tie formation.

4. Results

4.1. Textanalysis

To rate the tweets populism scores a system prompt was developed as described in Section 3.2 instructing the LLM with the operationalization scheme of populist rhetoric. To validate the prompt it was used to rate an already expert annotated dataset of sentences from bundestagspeeches used to train the PopBERT classifier (Erhard et al. 2025). The 8795 sentences were rated by 5 expert raters on a binary scale if they contain anti-elitist or people-centrist attitudes. since the LLMs system prompt rates the same dimensions in higher resolution the output was collapsed to a binary rating for comparison. The LLM rates if a tweet contains an attitude towards the elite or the people, each on a scale of -3 to 3 where negative numbers represent negative attitude towards the group and positive numbers positive attitudes. An elite_score from -3 to -1 represents anti-elitism and was collapsed to 1, while all other ratings were collapsed to 0. The same was done to the people_score.

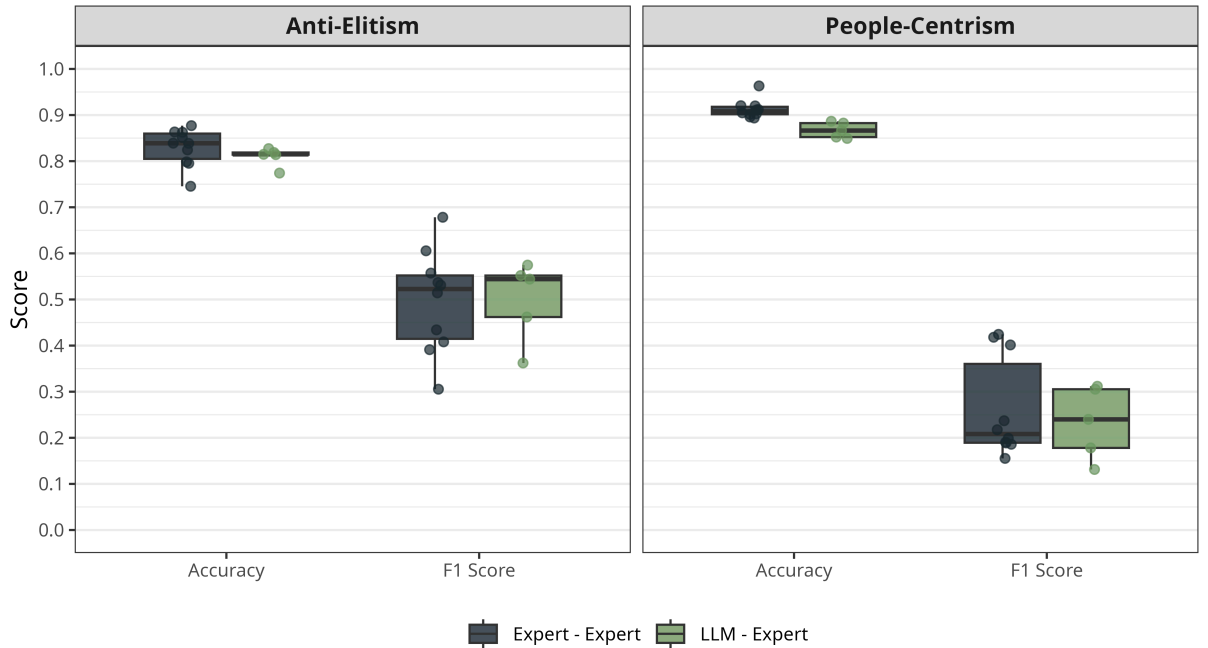


Figure 4: Prompt Performance - Pairwise accuracy and F1 scores for all expert-expert and LLM-expert rater combinations across anti-elitism and people-centrism.

Pairwise accuracy and F1 scores between binarized Qwen3-235B-A22B-Instruct-2507-FP8 ratings using the developed populism system prompt and five expert annotators from the PopBERT training corpus (Erhard et al. 2025). Each point represents one rater pair (expert-expert or LLM-expert). LLM ratings were collapsed from a 7-level Likert scale to binary scores (anti-elitism: score < 0 ; people-centrism: score > 0).

Figure 4 shows the pairwise accuracy measures and D1 scores between each combination of expert raters aswell as each combination between the LLM and one of the expert

raters. The rater-rater and rater-LLM accuracyscores are quite high for anti-elitism with a median between 0.8 and 0.9 and people-centrism with a median between 0.85 and 0.9. Accuracy is the proportion of tweets on which two raters assigned the same binary label. Since most sentences seemed to contain no anti-elitism or people-centrism the dataset was skewed which inflates the accuracy scores. The raters and the LLM have high agreement on the large count of 0 values.

To accurately assess the performance of the prompt pairwise F1-scores are calculated for each rater-rater and rater-LLM combination. It adjusts the accuracy by ignoring zero agreements and only rewards raters for agreement on rare positives. The rater-rater and rater-LLM combinations for the anti-elitism score produce a median F1 Score of slightly over 0.5, while the median F1 score for people-centrism lie around 0.2 - 0.25.

Populism seems to be a fuzzy concept where even expert raters have a hard time deciding on its presence on the anti-elitism dimension but agree even less on the people-centrism dimension. What still stands out is that the LLM-rater combinations lie in the same regime as the rater-rater combinations successfully making an LLM equipped with the systemprompt as precise as an expert rater in detecting dimensions of populism in bundestagspeeches.

The Qwen3-235B LLM rated all tweets by their attitude about the elite, the people and the opposition between the two groups.

Tweet	Populism Score	Party
Das kleben einem die politischen Freunde von @PoliticianX, sog. #Antifas an die Fenster, wenn sie nicht gerade Artikel von @PoliticianX in linksextremistischen Posttillen lesen. Die Aussage wird vom totalitären #innenministerium vermutlich geteilt. Das Volk soll die Fresse halten.	25	AfD
Prügelnde Bullen, Tränengas, Gewalt. Warum nochmal? Um die Verbreitung eines Krankheitserregers einzudämmen? Glaubt das ernsthaft noch jemand? Es ist längst klar, dass es hier nicht um Pandemiebekämpfung geht. Sondern um die 1% gegen den Rest. Steht auf und wehrt euch.	25	AfD
Und dieser transformierte demokratische Widerstand der Bürger wird am Ende wieder die Freiheit, die Demokratie und den Rechtsstaat herbeizwingen. "Lasst uns aus dem autoritären Linksstaat wieder einen freiheitlichen demokratischen Rechtsstaat machen!" #AfD	20	AfD
Typisch #Hannover, typisch unter #SPD-#Pistorius! #Polizei kuschelt mit #Clan-Kriminellen und #linksextremen Gegendemonstranten. Aber rein schikanösen #FFP2-#Maskenzwang (AN FRISCHER LUFT!) setzen Polizisten gegen friedliche Bürger brutal um, im Stil von "Waas guggst Du?"-Typen!	20	AfD
Gut erkannt! Aber das ist doch das Ziel, das immer währende Drangsalieren der Menschen. Durch #impfzwang #Impfpflicht Sog. #corona-Maßnahmen usw. Darum: #Spazierengehen! Dem gewissenlosen Establishment zeigen: #Impfpflichtneindanke	20	AfD

Table 1: Top-scoring tweets on the composite populism scale.

The individual dimensions are aggregated by user. By that each user is positioned in a three dimensional space of those values. Figure 5 visualizes this space. Most users lie

at the zero value on all axis (better seen in Figure 17). Most users lie in bottom half of the plot, tweeting against the elite. Of those users some are tweeting also against the people, but most tweet against the elite and for the people. Antagonism is getting stronger in the direction of the bottom corners of the space. The diagonal from the origin to the bottom right region is essentially the axis of populist operationalization. The distribution is skewed in this direction, visualizing the real populism signal. The Appendix includes a version of Figure 5 with the highlighted position of politicians.

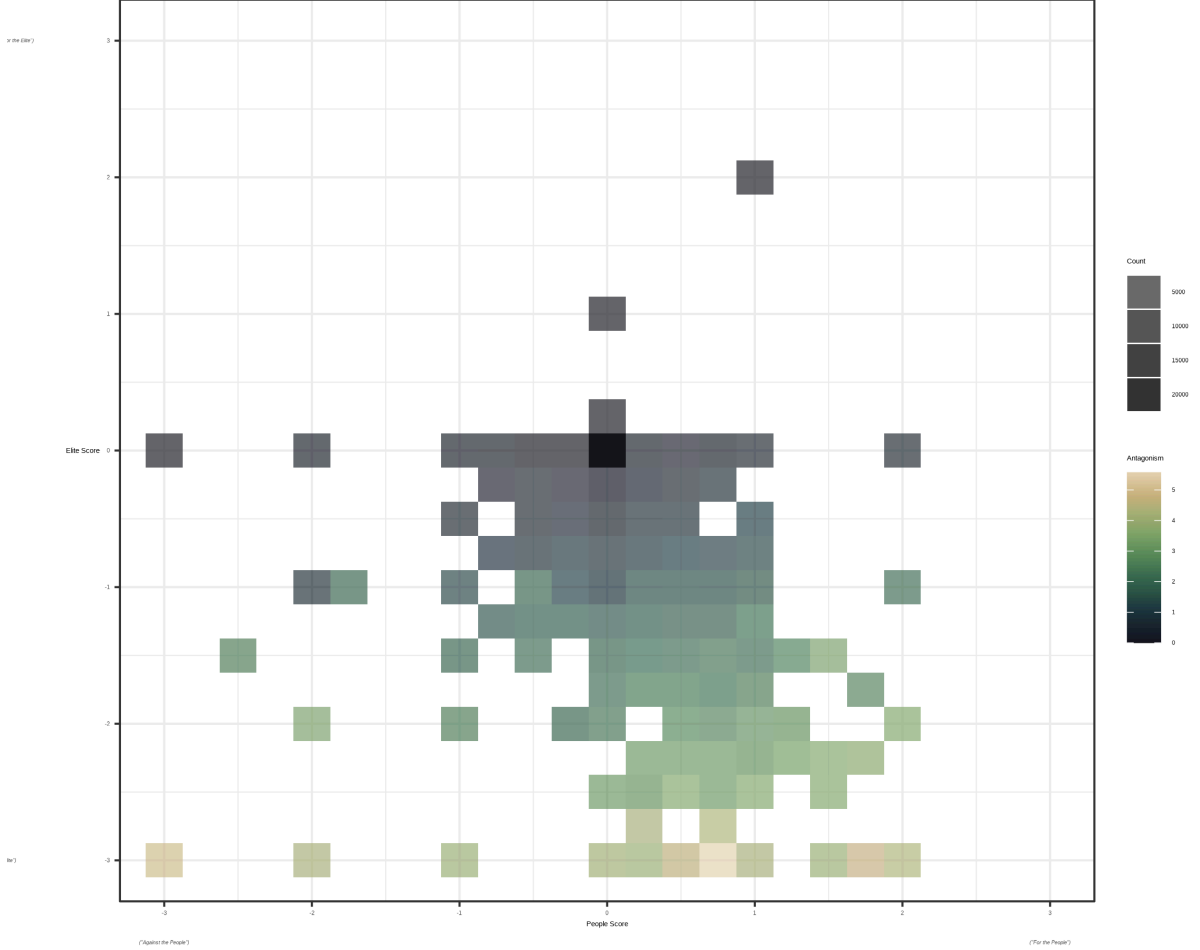


Figure 5: Distribution of populism dimensions per user in the giant component; People score ($\bar{p}_u$) and elite score ($\bar{e}_u$) are binned into 0.25-wide intervals. Color visualizes mean antagonism ($\bar{a}_u$) per bin; opacity describes user count ($n = 29\,672$).

Figure 6 shows the relative count of tweets rated on each scale, grouped by party (sorted by mean populism) and by politician within party (sorted by populism score relative to tweet count). The plot visualizes not only politicians of the giant component as before but all politicians in the original dataset including politicians outside of the giant component or politicians only present in the retweet network. This allows interpretation of within- and between-party variation in the populism score and its components. The AfD has the most politicians using populist rhetoric, followed by BSW and Die Linke, while CDU and CSU sit in the middle and the least populist parties are the Greens, SPD, and FDP which made up the ruling coalition at the time of data collection.

Populism scores are driven almost entirely by anti-elite and antagonism components, as pro-people rhetoric is nearly absent across all parties and therefore contributes little to the composite score. Most politicians do not use populist rhetoric on Twitter, except in the AfD, BSW, and Die Linke, where a substantial share use populism in at least 1/5 of their tweets. **Woah the skew in Figure 5 and party order in Figure 6 is external validation signal of the prompt method!!**

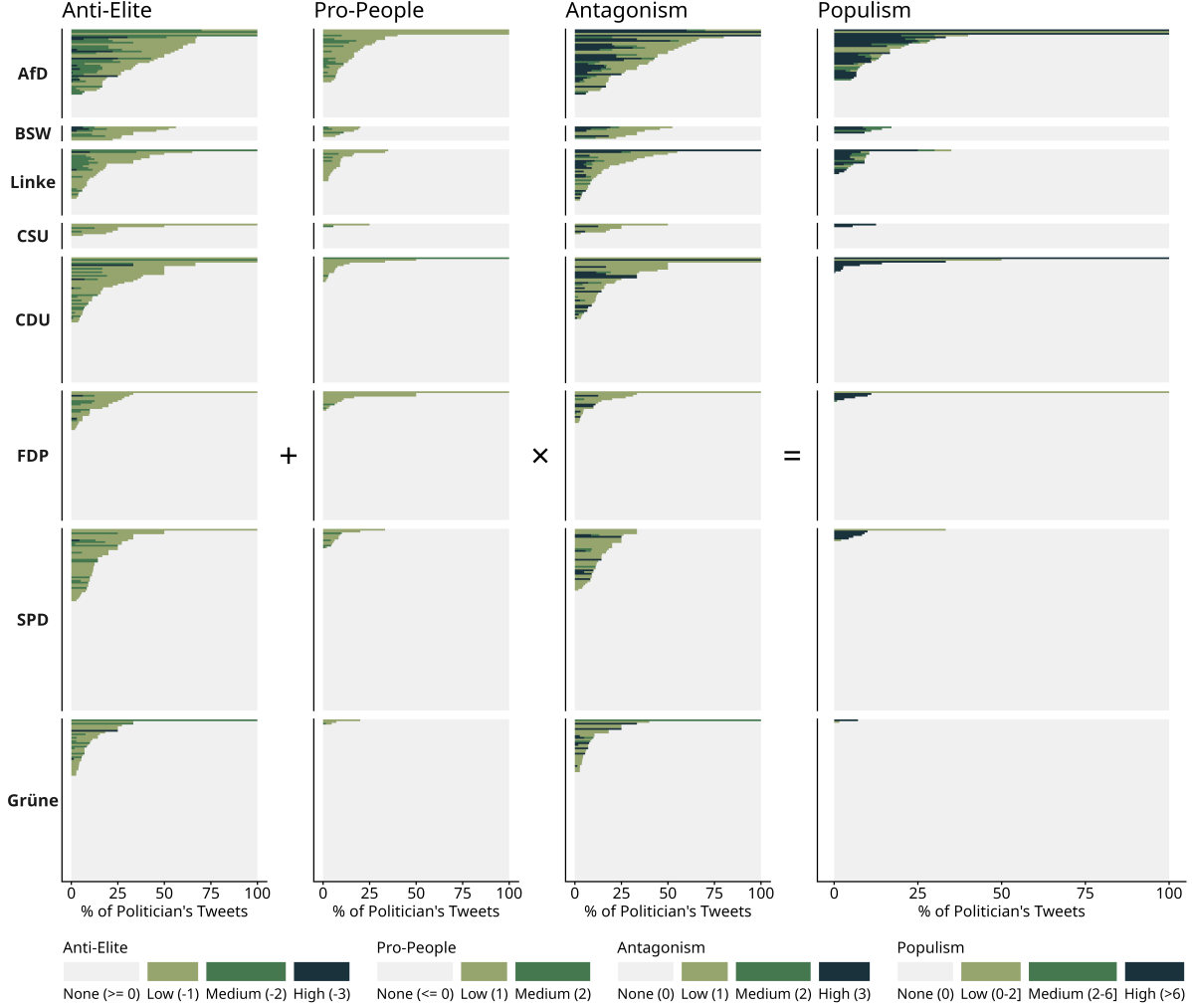


Figure 6: Populism score and its components split by parties and sorted by mean party populism scores.

Each bar is one politician; the x-axis shows the share of their tweets at each score level assigned by Qwen3-235B-A22B-Instruct-2507-FP8 with the expert level annotation prompt. The composite populism score equals $(\text{Pro People Score} + \text{Anti Elitism Score}) \times \text{Antagonism Score}$ when $\text{Antagonism Score} > 0$, otherwise $\text{Pro People Score} + \text{Anti Elitism Score}$. Dimensions visualized for direction which is increasing the populism score. Parties sorted by descending mean populism score; politicians sorted within party by non-zero share.

To get a deeper insight into the content of the tweets Figure 7 displays the term-frequency inverse-document-frequency correlation network colored by populism score. Tf-idf weights a term proportionally to its in-document frequency and inversely to its corpus-wide document frequency, treating concentration as a measure for distinctiveness. The top tfidf values across a corpus represent characteristic key words of the entire

The three populist categories share an actor and in-group rim of *volk*, *bürger*, *menschen*, *politiker*, *politik*, and *bevölkerung*, but the topical vocabulary they carry differs. Pro-People tweets foreground a normative-rights register (*freiheit*, *demokratie*, *staat*, *dürfen*, *bekommen*, *geld*, *millionen*) alongside pandemic terms (*corona*, *impfung*, *impfpflicht*, *maßnahmen*, *pandemie*, *lauterbach*). The arising topics are what citizens are owed and denied. Anti-Elite tweets shift to named opponents like parties (*cdu*, *fdp*, *grünen*, *spd*), *regierung*, and *lauterbach* cluster around *politiker* and *herr*, with the same pandemic vocabulary (*impfpflicht*, *maßnahmen*, *impfung*, *pandemie*) reframed as elite failure. Antagonistic tweets combine both parties, *regierung*, *lauterbach* and pandemic policy on one side while *bevölkerung*, *bürger*, *freiheit* and affective terms like *angst* on the other. This stages the people-elite confrontation directly. The non-populist (other) tweets show none of these patterns. The topics arising are about occurring political events and political procedures. (*ukraine*, *corona*, *impfung*, *impfpflicht*, *lauterbach*) appear but without any actor, surrounded by everyday vocabulary (*finde*, *gerne*, *vielleicht*, *sogar*, *eher*, *weniger*, *tweet*). Across the populist panels *volk* stands out with its high above 0.20 populist-score correlation. It is followed by the medium correlating *bürger*, *bevölkerung* and *politiker*.

4.2. Network Analysis

The analysis now turns to the structure of the reply network as the relational substrate on which politicians and citizens meet. Table 2 summarizes it as a directed weighted network anchored in politicians' threads as described in Section 3.3.

Metric	Value
Nodes	68240
Links	237 264
Density	0.0001
Degree - Mean	6.95
Degree - Standard Deviation	70.37
Reciprocity	0.13
Assortativity (Degree)	-0.10
Average Shortest Path (sampled)	4.35
Diameter (sampled)	47
Clustering Coefficient	0.1267
Max K-Core	24

Table 2: Descriptive Statistics of the reply network.

The reply network connects users with each other through weighted edges capturing the count of replies sent to each other. It is created from reply trees connecting tweets with each other when they reply to a original tweet or another reply of that original tweet.

As described in Section 3.3 only the giant component of the network is further analysed.

The giant component contains 68 240 users and 237 264 directed reply edges at a density of 0.0001, the expected baseline for largescale online networks where any given pair is overwhelmingly unlikely to be connected. Mean degree of 6.95 places the typical user at a handful of replies while the standard deviation of 70.37 is an order of magnitude larger, marking hubs that absorb and produce orders of magnitude more, structurally the politicians themselves. Reciprocity at 0.13 shows replies flow oneway: citizens reply at politicians, politicians rarely reply back. A clustering coefficient of 0.13 leaves local cohesion thin, two users replying to the same politician only seldomly form a tie themselves and triangles remain rare. Degree assortativity at -0.10 is negative, highdegree nodes connect to lowdegree nodes rather than to each other. Average shortest path of 4.35 against a diameter of 47 shows the hubs compress topological distance such that despite the sparsity any two users sit only a short reply chain apart, while the max kcore of 24 marks a small dense core embedded in an otherwise sparse periphery.

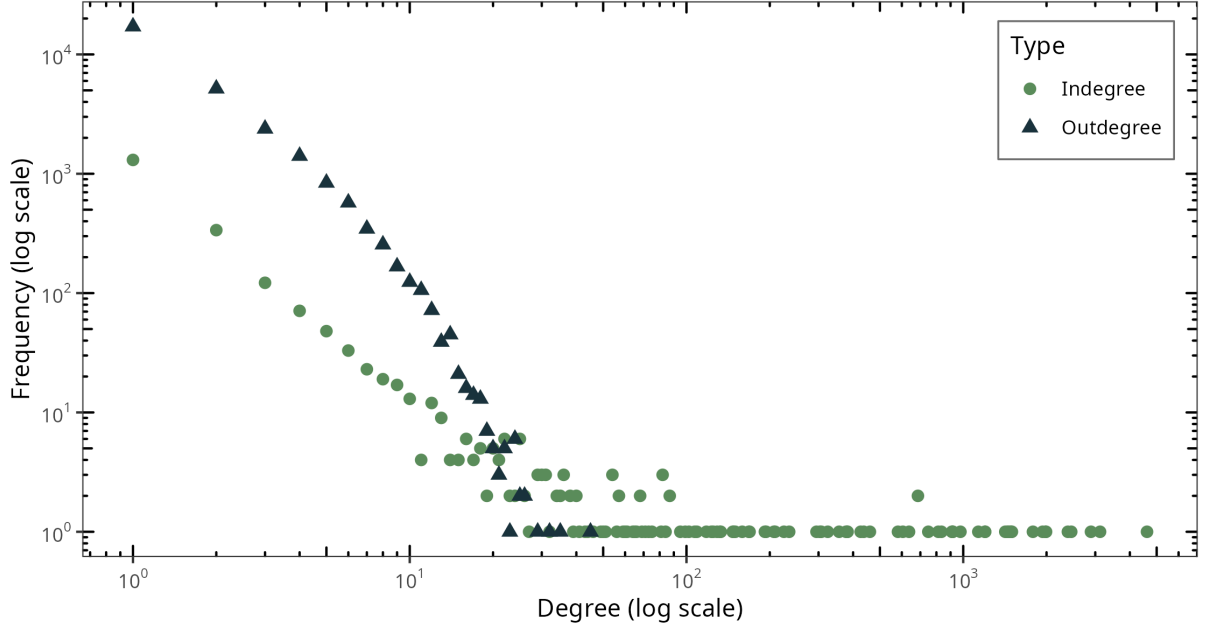


Figure 8: In and outdegree distributions of the giant component on log-log scale. Frequency of users with a given in or outdegree in the giant component ($n = 68\,240$). Indegree counts replies received per user, outdegree counts replies sent. Both axes log scaled. The outdegree distribution truncates at roughly 30 while the indegree tail extends past 10^3 , identifying politicians as the structural hubs of the reply network.

Figure 8 splits the degree distribution into its directed components and reveals two qualitatively different shapes that cross. Outdegree starts high at degree one with roughly $2 \cdot 10^4$ users sending a single reply, decays steeply across the loglog plane, and truncates near 30 with no user sending substantially more. Indegree starts an order of magnitude lower at degree one, decays more gradually, and the two curves cross around degree 10 to 20 where indegree overtakes outdegree and continues into a heavy tail reaching past 10^3 . In short the majority of users has a high outdegree and low

indegree, so post more than they receive, but only few users have a high indegree. This consolidates the picture from the Table 2.

To confirm the core-periphery network structure Figure 9 a k -core decomposition of the giant component peels the least connected nodes of the network network layer by layer, revealing how nodes distribute across the depth from periphery to dense core (Seidman 1983).

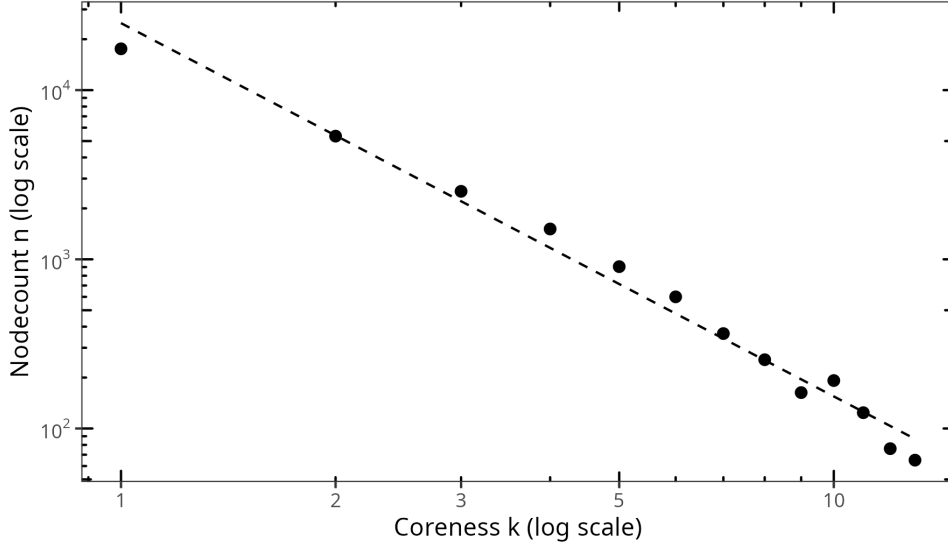


Figure 9: K-core decomposition of the giant component.

Nodecount n per coreness shell k for the giant component, on log-log axes. Each point gives the number of nodes whose maximum coreness equals k . The dashed line is a linear fit for reference. The approximately linear trend across more than three orders of magnitude indicates that shell sizes decay roughly as a power of k .

Shell sizes decline steeply and monotonically with coreness, spanning more than two orders of magnitude. The decomposition reveals a continuous gradient from a large periphery to a small dense core, without a sharp core/periphery boundary.

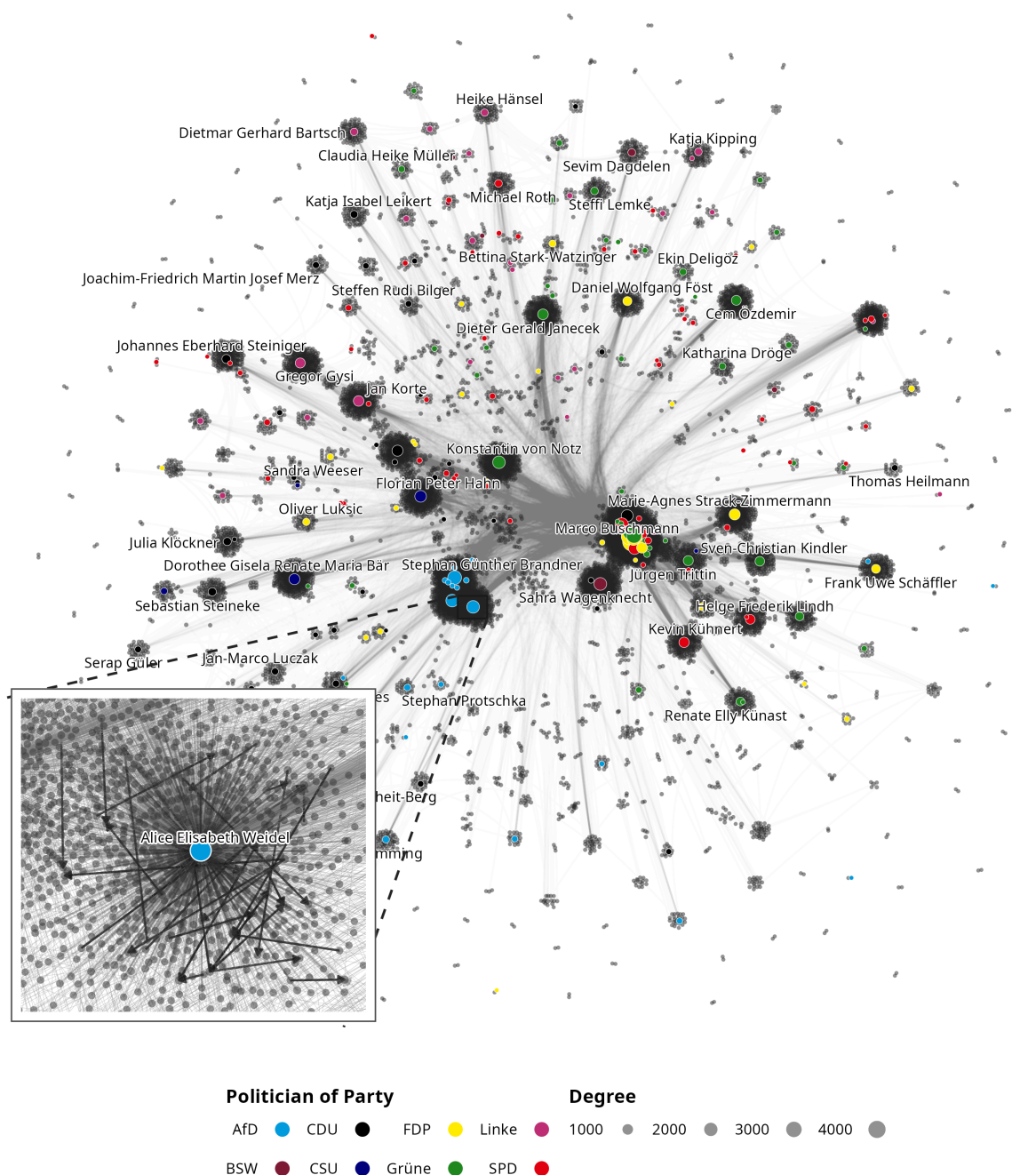


Figure 10: Giant Component of Reply Network
Reply Network layed out with thr DrL Algorithm (Martin et al. 2007). Nodesizes scale with degree, politicians are colored by their partycolor. The zoomed inset highlights the directed edges between alters in Alice Weidels' egonetwork.

Figure 10 visualises the giant component of the reply network. The layout is produced with the Distributed Recursive Layout (DrL) algorithm (Martin et al. 2007). At first a force directed network is layed out, which is then iteratively hierarchically clustered into fewer nodes until a sufficiently small network is created. In the end the finer graphs are drawn in again.

Structurally a comparatively small set of high-degree politicians functions as a dense core of the network. Each politician is surrounded by a close reply-community. Some of the communitymembers are then also active in other politicians reply networks connecting them with the core. It almost seems like an ideal hierarcical network in (Ravasz and Barabasi 2002)‘s sense, with the difference that this network has very low clustering (Table 2) due to broadcasting structure around each politician.

The hub-periphery structure is divided by a political boundary. Politicians of the parliamentary opposition (Die Linke, BSW, CDU/CSU, AfD) populate the left half of the figure, while members of the governing coalition (SPD, Grüne, FDP) cluster in the dense core and extend toward the right. Politicians whose audiences overlap with those of governing MPs are pulled into the centre, whereas those embedded primarily within partisan audiences are displaced toward the periphery, so that the layout itself recovers a coarse government–opposition distinction.

The inset zooms into Alice Weidel’s egonetwork and displays the broadcast pattern at the local level. The overwhelming majority of her alters are connected to her by direct replies to Weidel, but a non-trivial share of edges runs between alters themselves. This within-ego connectivity indicates that beyond pure dyadic broadcasting, smaller communities of repliers also interact with one another, hinting at community activity.

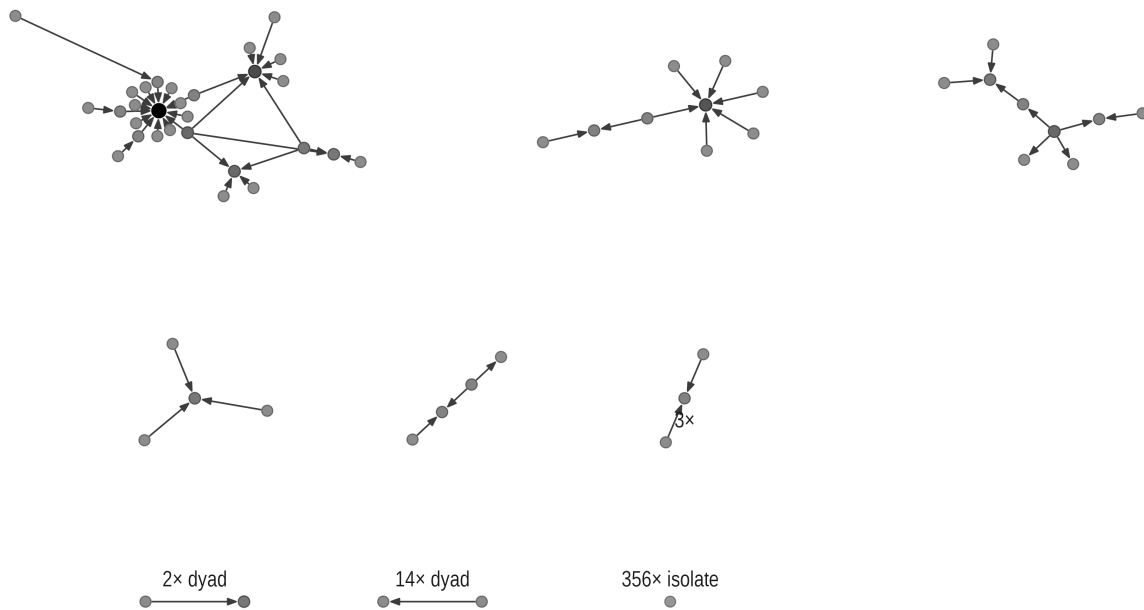


Figure 11: Ego Network - Alice Weidel

Egonetwork up to order 5. All reply-links created beneath other politicians original-tweets removed from a respective politicians egonetwork.

Figure 11 decomposes Alice Weidel’s egonetwork up to order 5 once all reply-links produced beneath other politicians original tweets have been pruned, so that only the ties formed within Weidel’s own conversational threads remain. The majority of Weidel’s

engagement-community do not interact with one another within her reply space. At the same time, the connected component and the handful of intermediate structures indicate some interaction patterns among the community. Remember, that all of the alters visualized also have replied to weidel at some point in the observed week.

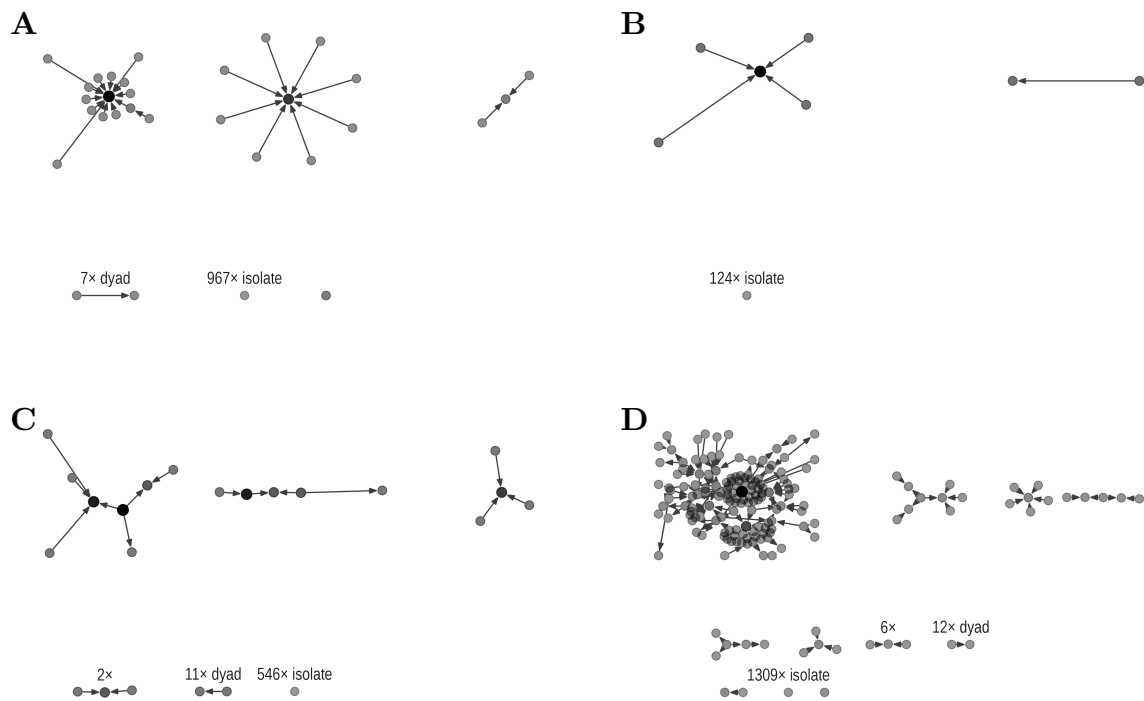


Figure 12: Ego Networks of four Politicians

Egonetworks up to order 5 for (A) Christian Lindner, (B) Katja Kipping, (C) Stephan Brandner and (D) Sahra Wagenknecht. All reply-links created beneath other politicians' original tweets have been removed from each respective politician's egonetwork.

4.3. Egonetwork Analysis

In order to investigate the connection between politicians populist language and the manifestation of the people ingroup in their respective engagement community regression models are estimated.

- why binarize populism
- explain model setup
 - The outcome variable is mean alter degree ...
 - discuss robustness check
 - control variables, what and why
- go interpret, go nuts
- do not forget table and checks in the appendix

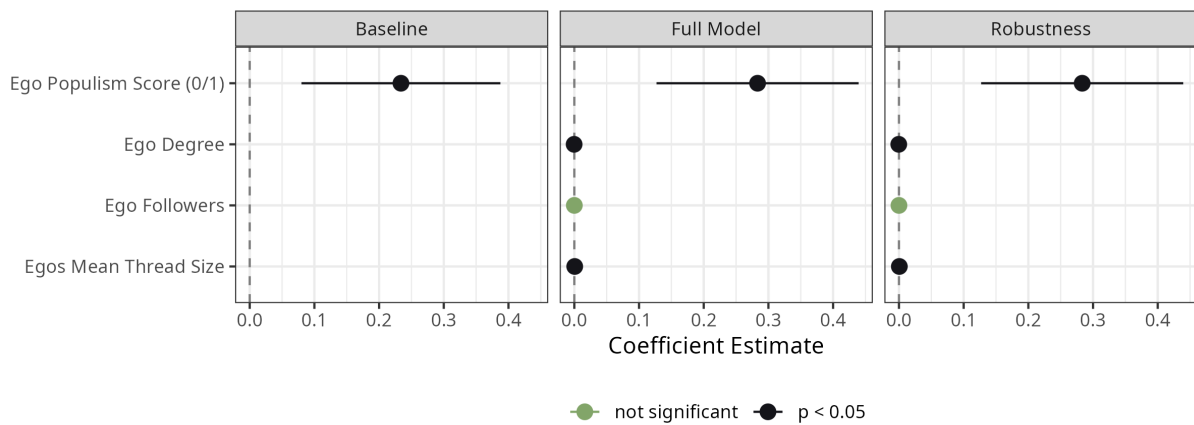


Figure 13: Estimating Mean Alter Degree with Politicians Populist Language

Predicted values

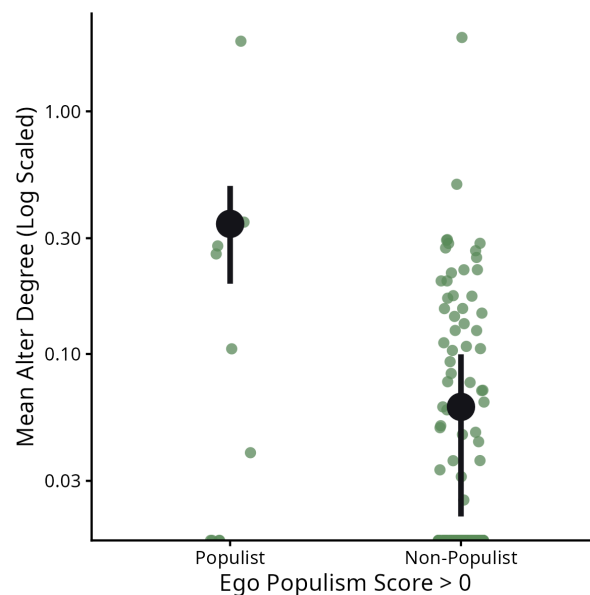


Figure 14:

Oh my god we have only useable egonetworks from X populist politicians of 147 in the giant component in general!!!

Skewed distribution does not allow a conclusive decision due to sparse data.

This is all great but this only controlled for ego variables. Is it possible to maybe distinguish the alter-alter connections created in populist vs. nonpopulist networks? Who replies to whom under politicians tweets?

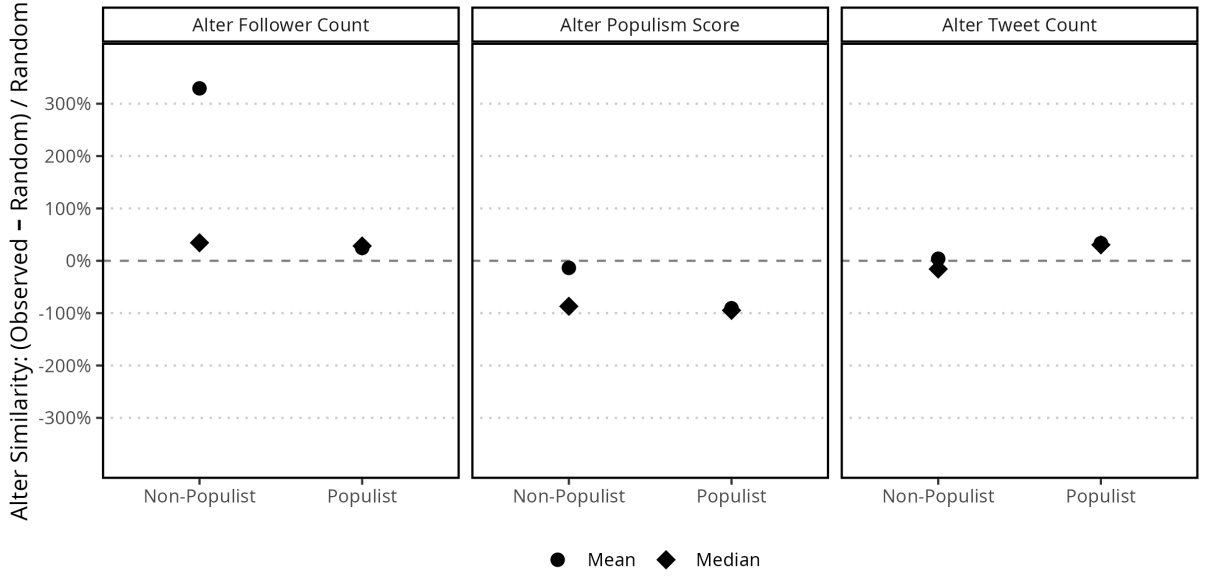


Figure 15: absdiff (Alter Similarity)

- What is shown
 - For each ego network, the mean absolute difference on a given attribute is computed once over all observed alter–alter ties and once over all possible alter–alter dyads. The ratio $(\text{observed} - \text{random}) / \text{random}$ yields one value per ego network. The circle shows the median across all ego networks in each group, the triangle shows the mean.
- Alter Follower Count
 - Medians are slightly positive for both groups (35% non-populist, 25% populist).
 - Connected alters tend to be more different in follower count than random pairs — mild heterophily.
 - The non-populist mean explodes to 330% while the median stays low, indicating a heavily right-skewed distribution driven by a handful of ego networks where very high- and very low-follower alters disproportionately connect. The populist distribution is far more compact (mean $\approx$ median).
- Alter Populism Score
 - The strongest signal. Both medians sit near -90% , meaning connected alters hold nearly identical populism scores compared to what random pairing would produce.
 - This is strong, consistent populism homophily across both network types.
 - The non-populist mean is pulled upward to roughly -15% , again indicating right skew from a few outlier ego networks; the populist mean and median nearly overlap, suggesting a tight, well-behaved distribution.
- Alter Tweet Count
 - Medians hover near zero for non-populists (-10%) and slightly positive for populists ($+30\%$).
 - There is no strong homophily or heterophily on tweet volume for non-populist ego networks, while populist ego networks show a mild tendency for connected alters to differ in tweet activity. Mean and median are close in both groups, so the distributions are relatively symmetric.

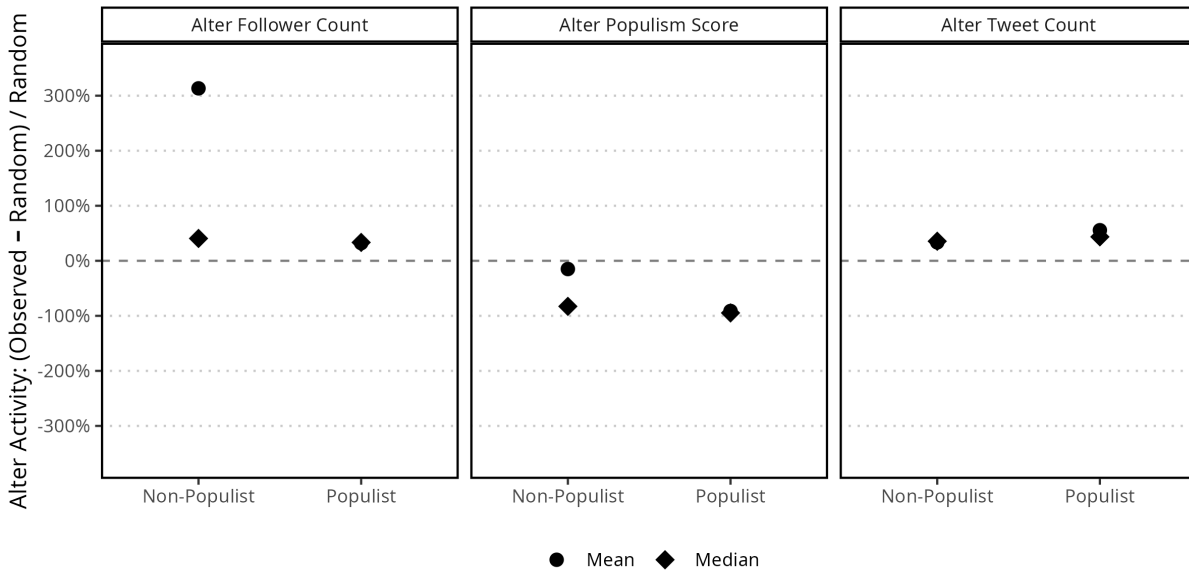


Figure 16: nodecov (Alter Activity)

- What is shown
 - Instead of the absolute difference, the sum of both alters' attribute values is computed per tie and per possible dyad. Positive deviation means ties preferentially form among alters with higher attribute values.
- Alter Follower Count
 - Both medians are positive (45% non-populist, 35% populist).
 - Alters with more followers connect to each other at higher-than-expected rates — a preferential attachment or popularity effect. The non-populist mean is again far above the median (315%), confirming the same small set of outlier ego networks seen in the absdiff plot.
- Alter Populism Score
 - Both medians are strongly negative (-80% to -90%), meaning ties form among alters with lower combined populism scores.
 - The more populist an alter pair is, the less likely they are to be connected. This mirrors the absdiff finding: connections cluster among ideologically similar, low-populism alters.
- Alter Tweet Count
 - Both groups are positive (medians 35% non-populist, 50% populist). High-volume tweeters connect more than expected.
 - The effect is somewhat stronger in populist ego networks. Mean and median are close in both groups, indicating stable distributions without extreme outliers.
- Summary
 - structure is the same across populist and non-populist ego networks!
 - strong populism homophily
 - mild follower heterophily
 - preferential connectivity among active and popular users

- ▶ degree difference
 - populist ego networks show slightly more tweet-driven heterophily and activity effects

but the qualitative signatures are remarkably similar. The recurring mean–median divergence in the follower facet for non-populist networks traces to a small number of high-visibility ego networks.

5. Discussion

6. Conclusion

Appendix A: Additional Tables, Figures and Equations

Symetry of F1 Score for swapping raters

Given binary ratings $a_i, b_i \in \{0, 1\}$ from two raters, define:

$$\text{both_positive} = \sum_i a_i b_i$$

$$\text{only}_{b_{\text{positive}}} = \sum_i (1 - a_i) b_i$$

$$\text{only}_{a_{\text{positive}}} = \sum_i a_i (1 - b_i)$$

Treating A as ground truth:

$$\text{precision} = \frac{\text{both_positive}}{\text{both_positive} + \text{only}_{b_{\text{positive}}}}, \quad \text{recall} = \frac{\text{both_positive}}{\text{both_positive} + \text{only}_{a_{\text{positive}}}}$$

Swapping roles swaps $\text{only}_{a_{\text{positive}}} \leftrightarrow \text{only}_{b_{\text{positive}}}$, which swaps precision and recall. Since F_1 is their harmonic mean:

$$F_1 = \frac{2 \cdot \text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} = \frac{2 \cdot \text{recall} \cdot \text{precision}}{\text{recall} + \text{precision}}$$

Multiplication and addition are commutative, so F_1 is invariant under the swap. ■

For swapping labels (0,1) F Scores are not symetric.

Component Tables

Component Size	Number of Components	Total Nodes	% of Network
77194	1	77194	94.96%
41	1	41	0.05%
34	1	34	0.04%
31	1	31	0.04%
27	1	27	0.03%
23	1	23	0.03%
19	1	19	0.02%
16	1	16	0.02%
14	1	14	0.02%
11	3	33	0.04%
10	2	20	0.02%
9	4	36	0.04%
8	6	48	0.06%
7	7	49	0.06%
6	13	78	0.10%
5	24	120	0.15%
4	46	184	0.23%
3	185	555	0.68%
2	1207	2414	2.97%
1	359	359	0.44%

Table 4: Component size distribution of the reply network linking users through directed and weighted reply links. A giant component holds most user nodes and the size appearing the most is two.

Textanalysis Additional Results

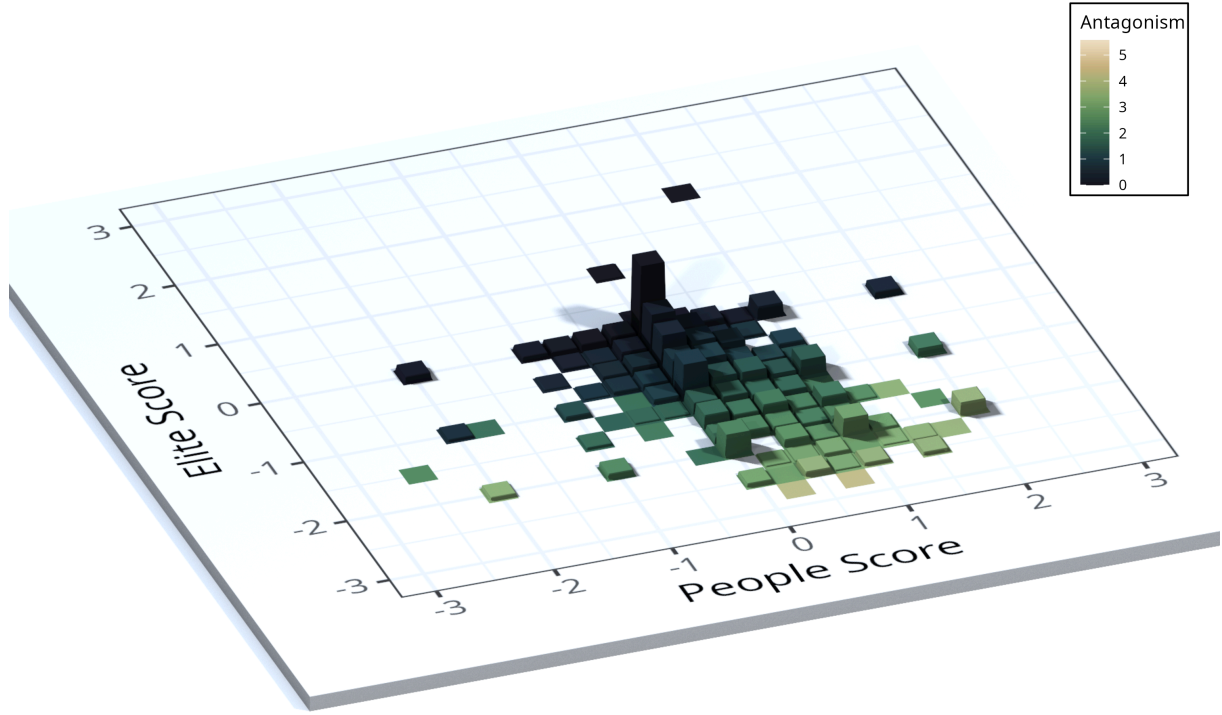


Figure 17: Distribution of populism dimensions per user in the giant component; People score ($\bar{p}_u$) and elite score ($\bar{e}_u$) are binned into 0.25-wide intervals. Color visualizes mean antagonism ($\bar{a}_u$) per bin; height describes user count ($n = 29\,672$).

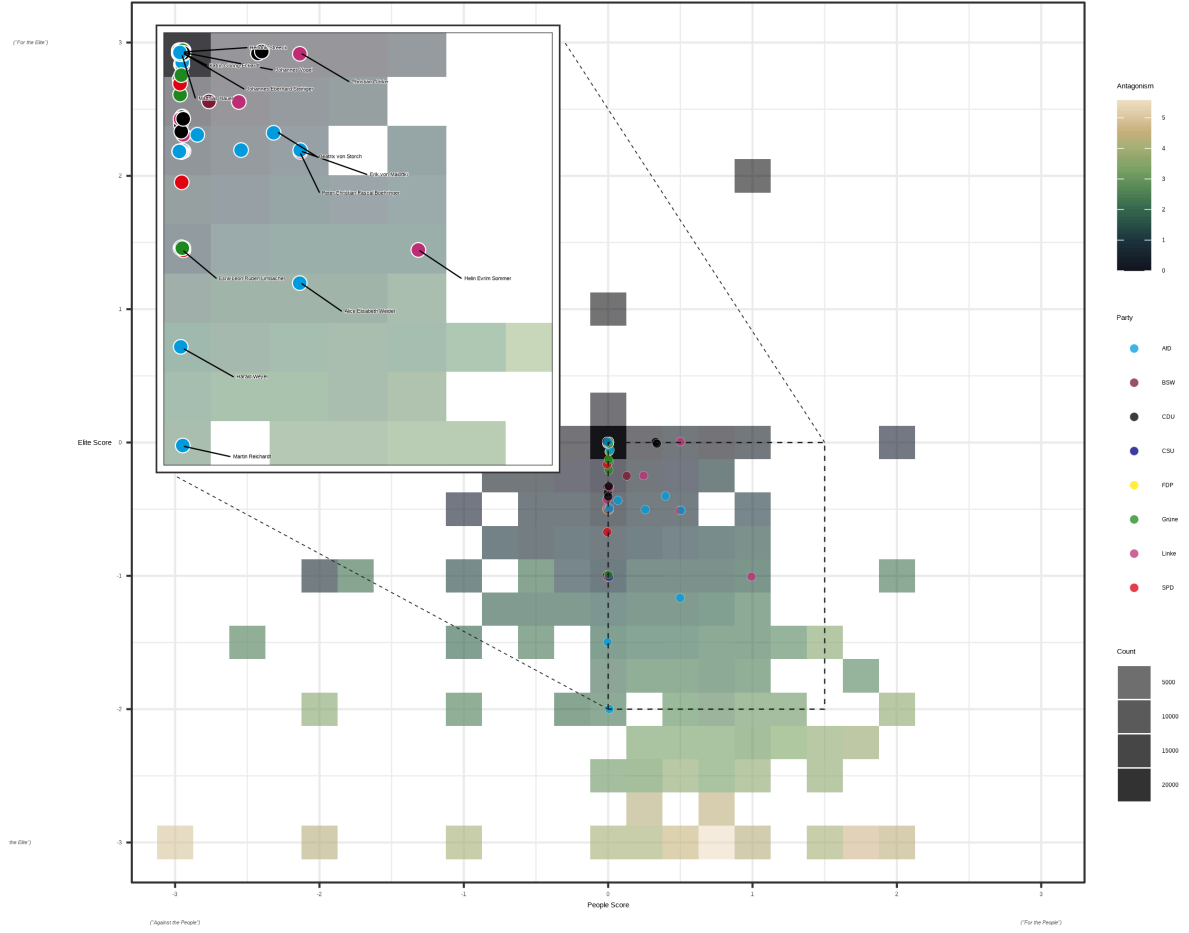


Figure 18: Distribution of populism dimensions per user in the giant component with marked politicians position;

People score ($\bar{p}_u$) and elite score ($\bar{e}_u$) are binned into 0.25-wide intervals. Color visualizes mean antagonism ($\bar{a}_u$) per bin; opacity describes user count ($n = 29\,672$).

Appendix C: Prompt

You are a political sociology analyst trained to detect populist rhetorical strategies in social media posts. Your goal is to identify which posts contain dimensions of populist attitudes – without judging political views.

Analyze each post holistically. Avoid speculation.

Pre-Analysis Check

Before analyzing, ask: ****Does this text carry a detectable rhetorical signal**** – through language, symbols, or clear implication?

If you find no signal at all (e.g., only user mentions, hashtags, URLs, or emojis with no interpretable stance), return the JSON with all scores as 0 and the holistic redescription as: "No semantic argument detected."

If the signal is ambiguous but present, proceed with analysis and flag uncertainty.

Dimensions of Populism

1. ****People Attitude (The Ingroup)****

- ****Definition:**** Arguing in favour (positive attitude) or against (negative attitude) a large homogenous ingroup ("The People") considered the societal norm.
- "The People" must imply the ****universal ordinary majority**** – e.g., "The common people," "The citizens," "We," "Us," "This country's people."
- A single individual or a list of named individuals is not "The People."
- Specific subgroups (e.g., "families," "pensioners," "minorities") are not "The People" on their own.

2. ****Elitist Attitude (The Outgroup)****

- ****Definition:**** Arguing in favour (positive attitude) or against (negative attitude) a small, powerful outgroup.
- "The Elite" refers to a powerful class or group – e.g., "politicians," "the ruling class," "the establishment," "the media."

3. ****Antagonism (The Divide)****

- ****Definition:**** Antagonism captures the strength of opposition between The People and The Elite.

Analysis Instructions

Step 1: Holistic Redescription** (concise – aim for roughly one paragraph)

- Synthesize the post's strategic purpose:
 1. Summarize the core narrative.
 2. Identify the rhetorical structure.

Step 2: Actor & Signal Analysis

Identify every person or group referenced, then work through these questions for each:

1. ****Scale:**** Is this actor an individual, a named role or institution, or a generalized class? *_How confident am I – what would the alternative reading be?_*
2. ****People signal:**** Does the text invoke a broad ordinary majority? *_Or am I inferring "The People" just because an elite is attacked? If the text names a specific subgroup rather than "The People" as a whole: does the text frame this subgroup as representing the ordinary majority's experience, or is it advocating for that group's particular interests?_*
3. ****Elite signal:**** Does the text target a powerful group or class? *_Or is it criticizing a specific policy or a single person without generalizing?_*
4. ****Antagonism signal:**** Does the text construct opposition between people and elite? *_How strong is the moral charge – is the elite merely wrong, or morally culpable, or an existential threat?_*

If a reasonable coder could disagree with your classification on any of these, state the alternative interpretation and flag your confidence as LOW. Otherwise, flag HIGH.

Step 3: Scoring

Score each dimension using the scales below.

****People Attitude (The Ingroup)****

```

- ++3: "The People" are viewed as virtuous or superior.
- ++2: Clear support for "The People."
- ++1: Mild support.
- 0: Neutral or absent.
- -1: Mild criticism of "The People."
- -2: Clear criticism of "The People."
- -3: "The People" are viewed as incompetent or irrational.

**Elitist Attitude (The Outgroup)**
- ++3: "The Elite" is viewed as virtuous or superior.
- ++2: Clear support for "The Elite."
- ++1: Mild support.
- 0: Neutral or absent.
- -1: Mild criticism of "The Elite."
- -2: Clear criticism of "The Elite."
- -3: "The Elite" are viewed as incompetent or irrational.

**Antagonism (The Divide)**
- 0 (No divide): No opposition is framed between the people and the elite.
- 1-2 (Dissatisfaction): Passive complaints or implied asymmetry without a moral charge.
- 3-4 (Active blame): The elite is framed as responsible and culpable; characterized by insults, accusations of dishonesty, and explicit us-vs-them framing.
- 5-6 (Existential threat): The elite is framed as an enemy; includes dehumanization, conspiratorial framing, and calls for collective action or eliminationist language.

```

Few-Shot Examples: Populist Rhetoric Analysis

Example 1: Strong Populist Signal

```

_INPUT TEXT TO ANALYZE:
> "The hardworking people of this country are sick and tired of corrupt politicians lining their pockets while ordinary families can't afford groceries. It's time WE take our country back from the Washington elites who have rigged the system against us. They don't care about you – they never did."

_OUTPUT:
```json
{
 "holistic_redescription": "The post constructs a classic populist narrative pitting a virtuous, suffering ordinary majority ('the hardworking people,' 'ordinary families,' 'WE,' 'us,' 'you') against a morally corrupt political elite ('corrupt politicians,' 'Washington elites'). The rhetorical structure frames the elite as deliberately exploitative ('lining their pockets,' 'rigged the system') and emotionally indifferent ('they don't care about you'), while issuing a mobilizing call to collective action ('take our country back'). The moral charge is strong: the elite is not merely incompetent but actively culpable.",
 "actor_analysis": "1. 'The hardworking people of this country' / 'ordinary families' / 'WE' / 'us' / 'you' – generalized class representing the broad ordinary majority. Confidence: HIGH. 2. 'Corrupt politicians' / 'Washington elites' – generalized class representing a powerful political outgroup. Confidence: HIGH. No individual is named; the criticism targets the political class as a whole.",
 "people_explanation": "A broad ordinary majority is explicitly and repeatedly invoked through multiple framings: 'hardworking people of this country,' 'ordinary families,' and inclusive pronouns ('WE,' 'us,' 'you'). These are not specific subgroups advocating particular interests – they represent the universal ordinary citizen. The People are framed as virtuous ('hardworking') and suffering ('can't afford groceries').",
 "people_score": 2,
 "elite_explanation": "A powerful political class is explicitly targeted as a generalized outgroup – 'corrupt politicians' and 'Washington elites.' The criticism is not directed at a single individual or a specific policy but at the political establishment as a whole. They are characterized as morally corrupt ('lining their pockets'), deliberately manipulative ('rigged the system'), and callously indifferent ('don't care about you').",
 "elite_score": -3,
 "antagonism_explanation": "The text constructs sharp, moralized opposition between The People and The Elite. The elite is framed as actively culpable – they have 'rigged the system' and are 'corrupt.' The language escalates beyond dissatisfaction into active blame with a mobilizing call ('take our country back'), suggesting the elite's control is near-illegitimate. The moral charge is strong (culpable and indifferent), though the post stops short of dehumanization or conspiratorial framing, placing it in the upper range of active blame.",
 "antagonism_score": 4
}
```

```

Example 2: No Detectable Populist Signal

```

_INPUT TEXT TO ANALYZE:
> "Just finished watching the new season of that baking show 🍰 Honestly episode 4 was wild. Anyone else think Marco should have stayed longer? #BakeOff #TeamMarco"

_OUTPUT:

```

```

```json
{
 "holistic_redescription": "No semantic argument detected.",
 "actor_analysis": "One individual is mentioned ('Marco'), who is a contestant on a television show. No political actors, institutions, generalized classes, or power dynamics are referenced. Confidence: HIGH.",
 "people_explanation": "No broad ordinary majority is invoked. The post is a personal entertainment opinion with no political or societal framing.",
 "people_score": 0,
 "elite_explanation": "No powerful group or class is referenced or targeted. The post contains no political or institutional criticism.",
 "elite_score": 0,
 "antagonism_explanation": "No opposition between any groups is constructed. The post is entirely about a television program.",
 "antagonism_score": 0
}
```

```

****Example 3: Ambiguous / Mild Signal****

INPUT TEXT TO ANALYZE:

> "Another satisfisatisfying satisfying satisfying satisfying day of the government announcing new taxes while somehow never fixing the roads or the schools. But sure, keep telling us everything is fine. We're not blind."

OUTPUT:

```

```json
{
 "holistic_redescription": "The post expresses frustration with government performance, using sarcasm ('But sure, keep telling us everything is fine') to highlight a perceived gap between government claims and lived reality. The rhetorical structure implies a contrast between an underserved public ('us,' 'we') and an ineffective government, but the framing remains at the level of policy dissatisfaction – criticizing specific failures (roads, schools, taxes) – rather than constructing a full populist morality tale. The 'us' vs. government framing is present but mild.",
 "actor_analysis": "1. 'The government' – a named institution, not a generalized elite class. However, 'the government' could be read as standing in for the broader political establishment. Confidence: LOW – a reasonable coder could interpret this as either targeted institutional criticism or broader anti-elite sentiment. 2. 'Us' / 'We' – an inclusive pronoun suggesting the general public. Could represent the broad ordinary majority or simply the speaker and their community. Confidence: LOW – the referent is ambiguous between a universal 'people' and a more local or personal 'us.'",
 "people_explanation": "The pronouns 'us' and 'we' suggest a broad public, and 'We're not blind' implies the ordinary citizen sees through official narratives. However, the post does not explicitly frame 'The People' as a virtuous or unified majority – it is closer to a shared complaint from a general public perspective. The signal is present but mild, and a coder could reasonably score this as 0 (just a personal complaint) or +1 (mild invocation of ordinary people's experience).",
 "people_score": 1,
 "elite_explanation": "The criticism targets 'the government' as an institution, focusing on specific policy failures (taxes, roads, schools) and dishonesty ('telling us everything is fine'). This sits between criticizing a specific institution's performance and painting a broader elite as out of touch. Because 'the government' is a single institution and the complaints are policy-specific, this leans toward institutional critique rather than generalized anti-elite framing. Score reflects mild criticism of a powerful institution without full generalization to an elite class.",
 "elite_score": -1,
 "antagonism_explanation": "There is an implied asymmetry: the government collects taxes but fails to deliver, and it dismisses the public's awareness ('telling us everything is fine' vs. 'we're not blind'). This constructs a mild divide – the government is framed as dismissive and ineffective, and the public is positioned as the aware, underserved party. However, the moral charge is limited: the government is incompetent and possibly dishonest, but not painted as existentially threatening or conspiratorially corrupt. This fits the dissatisfaction range.",
 "antagonism_score": 2
}
```

```

Output Format

Only answer with a codeblock containing a JSON file in the following format:

```

```json
{
 "holistic_redescription": "Synthesize the post's core narrative and rhetorical structure (roughly one paragraph).",
 "actor_analysis": "List all actors. For each: individual, role/institution, or generalized class? State confidence (HIGH/LOW) and note alternative readings where applicable.",
 "people_explanation": "Is a broad ordinary majority invoked – or are you inferring 'The People' from other signals? If a subgroup is named, does it stand in for the majority or represent particular interests?",
 "people_score": 0,
 "elite_explanation": "Is a powerful group or class targeted – or is the text criticizing a specific policy or individual without generalizing?",
 "elite_score": 0,
 "antagonism_explanation": "Does the text construct opposition between people and elite? How strong is the moral charge – merely wrong, culpable, or existential threat?",
 "antagonism_score": 0
}
```

```

}
, , ,

INPUT TEXT TO ANALYZE:

Appendix B: Code

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